DOCUMENT CONTROL & METADATA			
Project Name:	Thermal Analysis for Heat Exchanger Baffler Type	**Lead Analyst:**	AVX Engineering Team
Simulation Type:	Thermal Analysis	**Software Tool:**	Ansys 2024
Revision:	v1.0	**Date**	Oct. 08, 2026

1. Simulation Setup

Ansys Fluent	
Analysis Type	Internal
Unit System	SI (m-kg-s)
Fluid	Water
Total Cell Counts Mesh	807,555 & 1,144,295
Flow Type	Laminar and Turbulent
Gravity	On
Viscous	k-epsilon
Calculation Time	153 hours

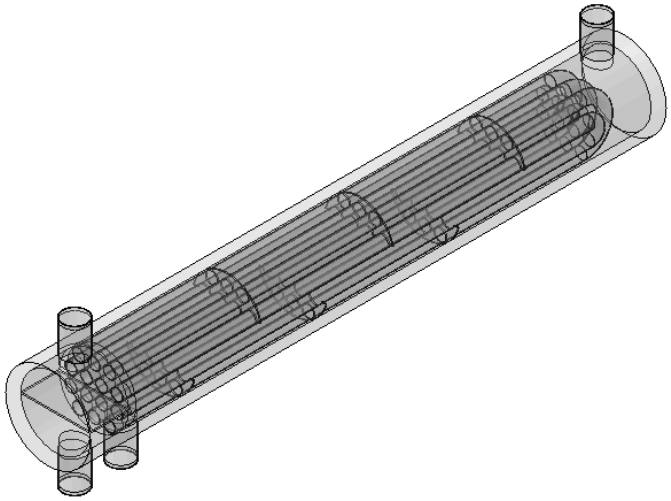


Figure 1. Circular Tube SolidWorks 3D Model

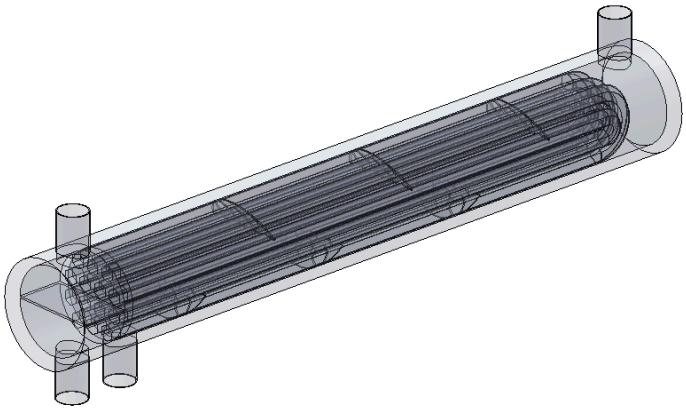


Figure 2. Hexagon Tube SolidWorks 3D Model

2. Simulation Report

Circular Mesh Independence Study			
Mesh Size	**Mesh Element Counts**	**Result**	**%**
0.6	359394	20133.945	
0.461538462	627146	18718.742	7.560353148
0.355029586	**1144295**	**18031.883**	**3.809136295**
0.273099681	2156210	17562.537	2.67242711

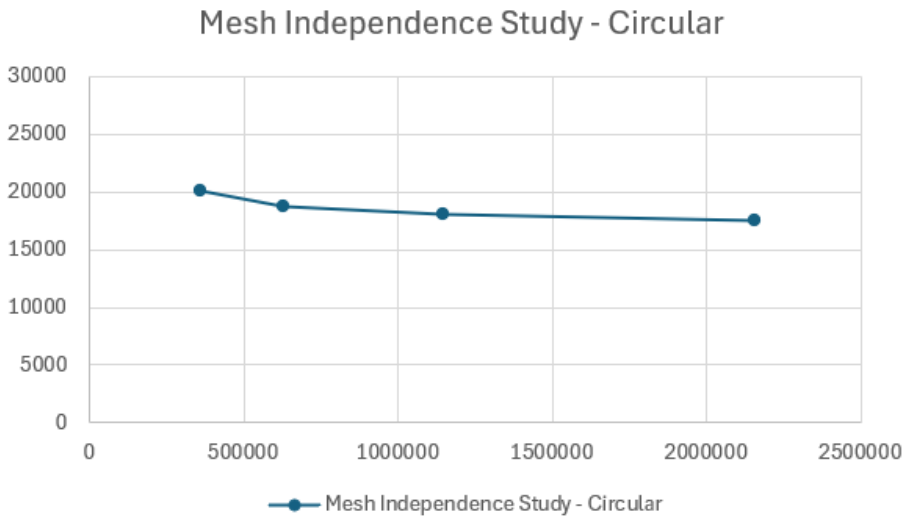


Figure 3. Circular Tube Mesh Independence Study

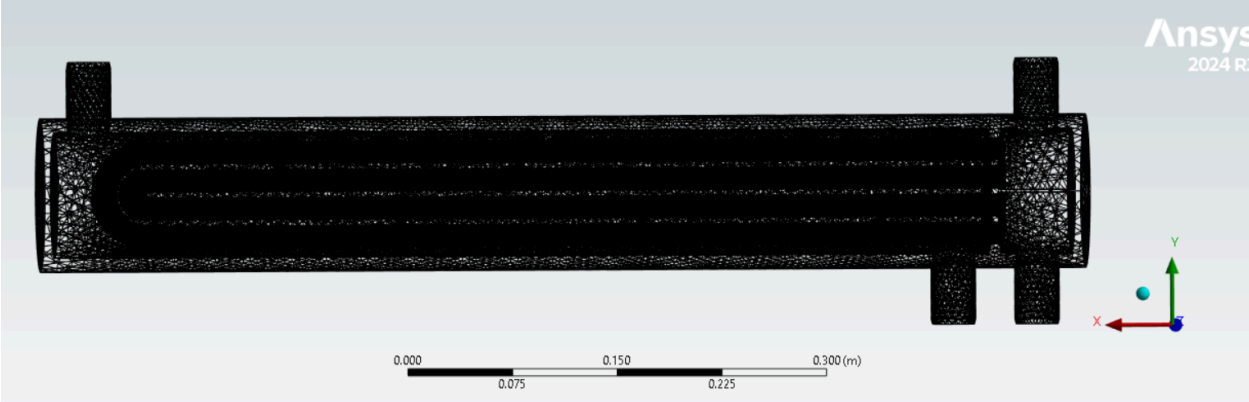


Figure 4. Circular Tube Mesh - 1,144,295 cells

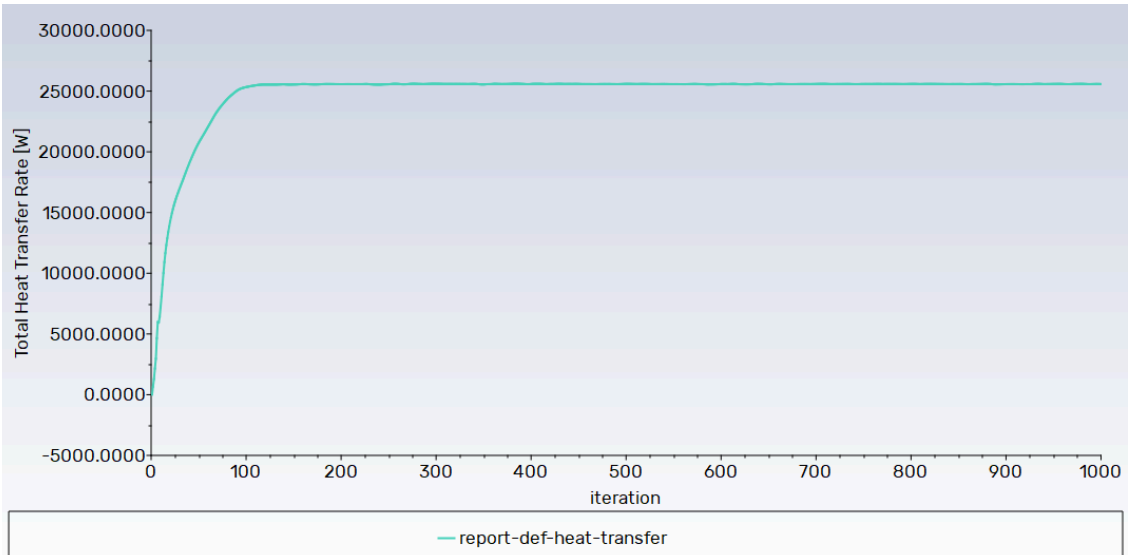


Figure 5. Heat Transfer Convergence Result - Circular

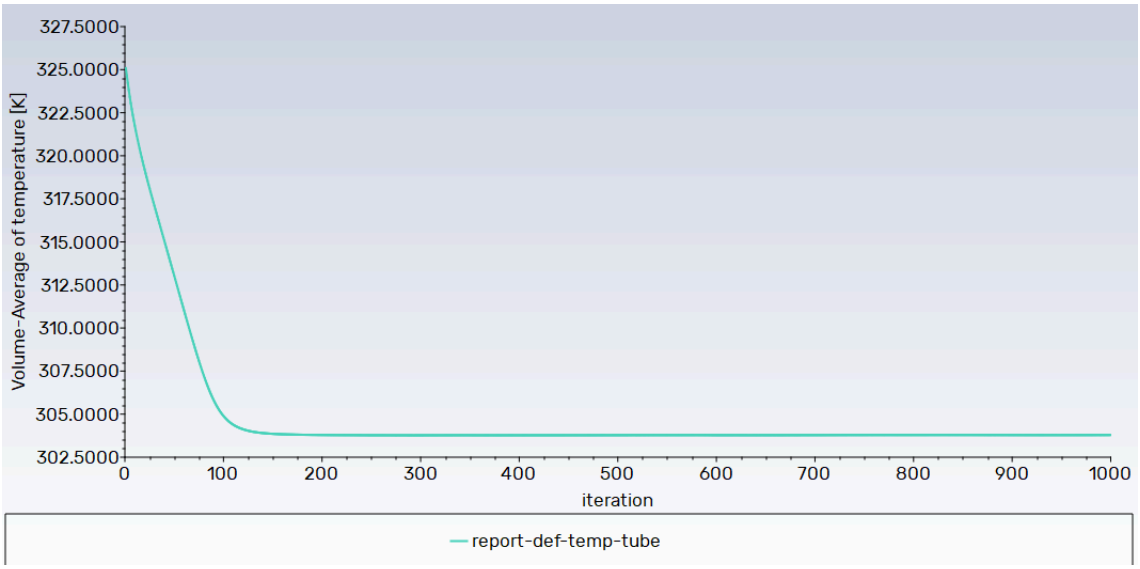


Figure 6. Temperature Tube Convergence Result - Circular

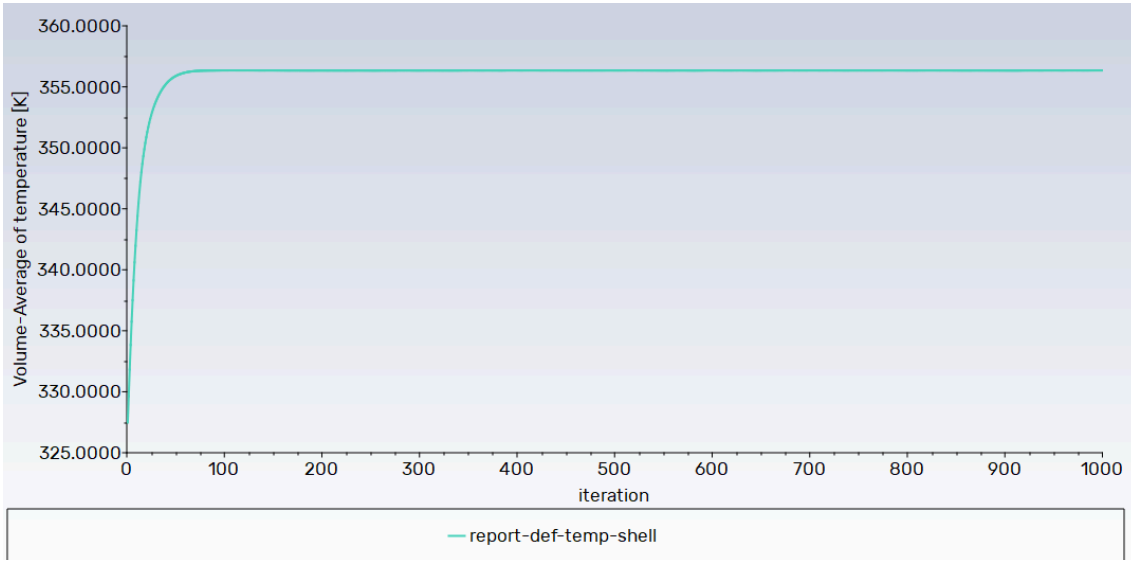


Figure 7. Temperature Shell Convergence Result - Circular

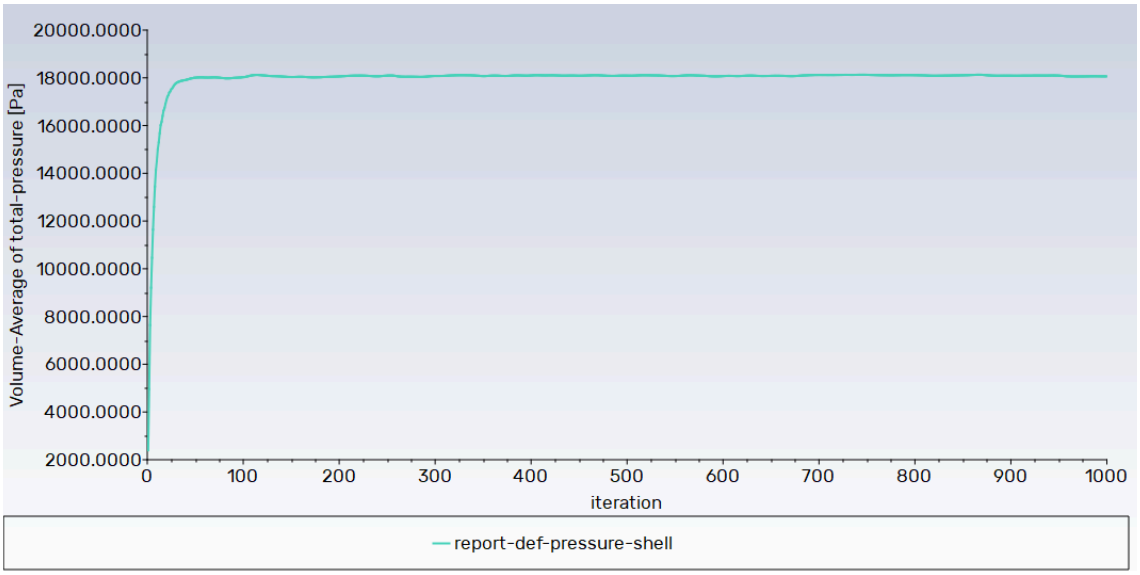


Figure 8. Pressure Shell Convergence Result - Circular

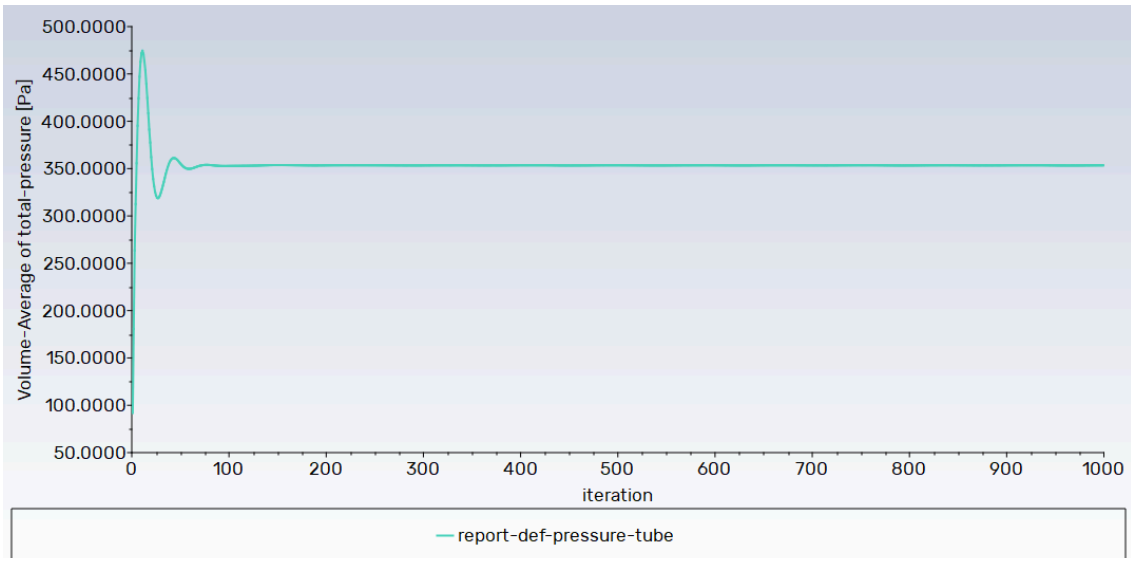


Figure 9. Pressure Tube Convergence Result - Circular

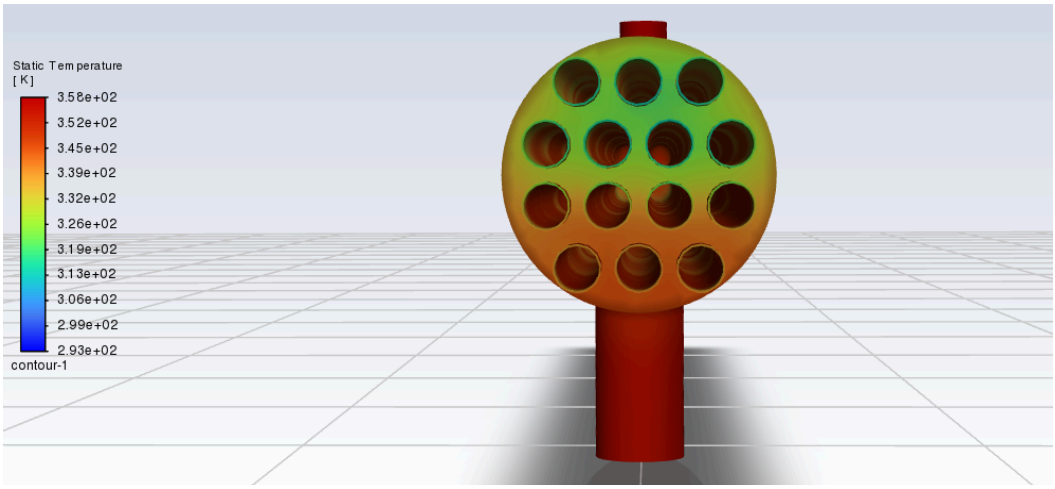


Figure 10. Post-Process View 1 - Circular

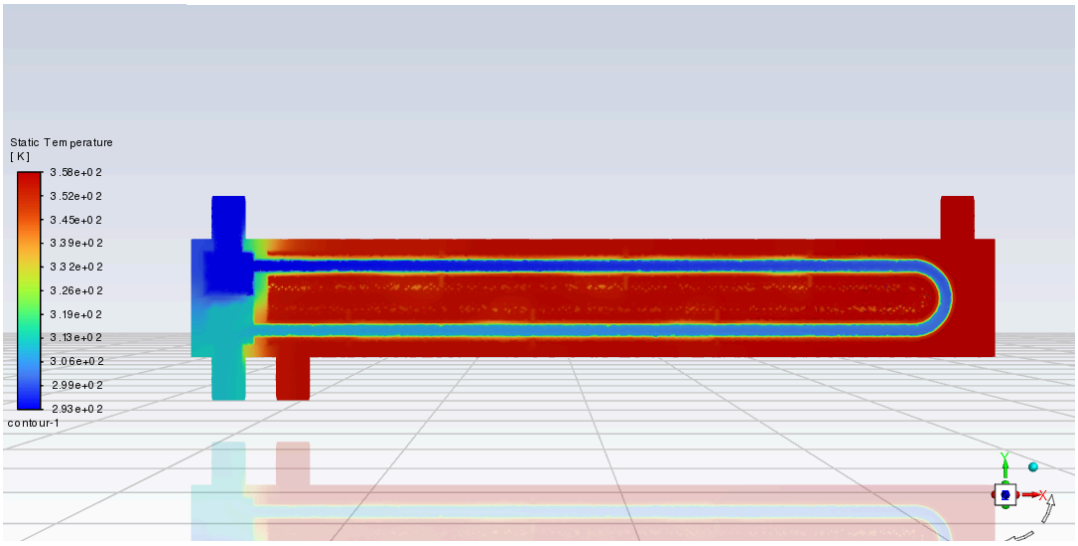


Figure 11. Post-Process View 2 - Circular

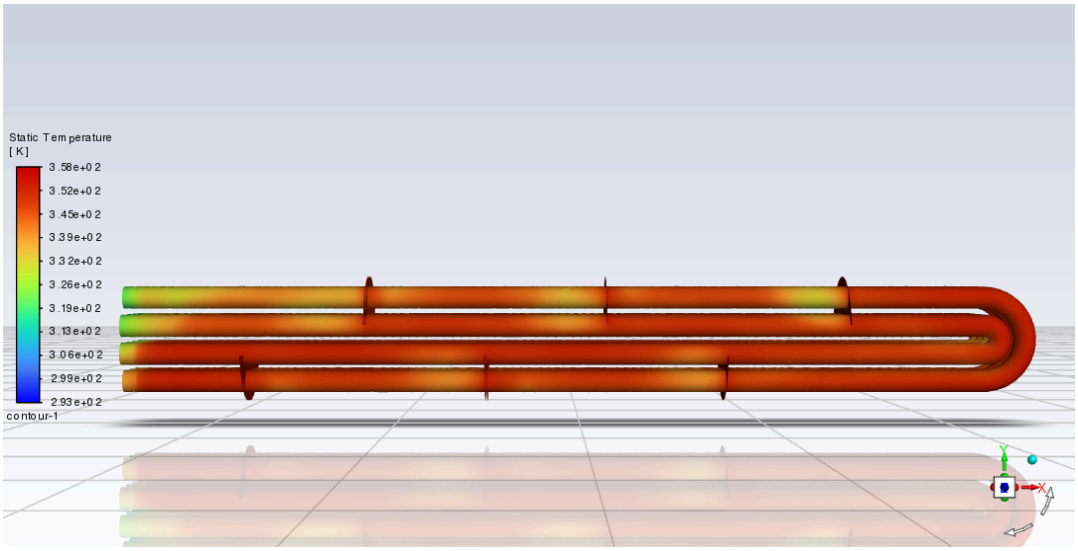


Figure 12. Post-Process View 3 - Circular

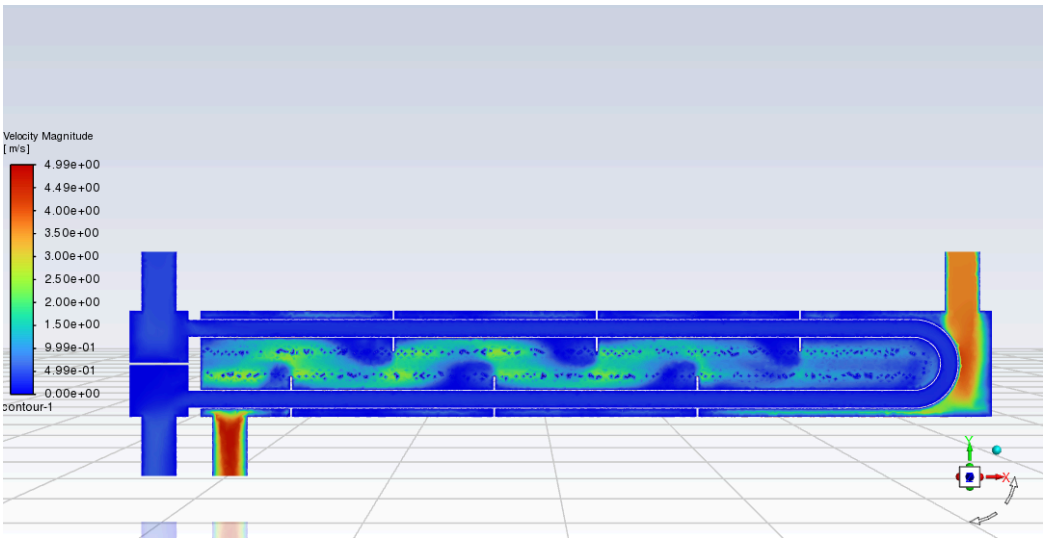


Figure 13. Post-Process View 4 - Circular

Circular Tube Results	
Heat Transfer	**25,579.57 W**
Temperature Tube	**303.78 K**
Temperature Shell	**356.34 K**

Circular Tube Results	
Pressure Shell	**18,061.43 Pa**
Pressure Tube	**353.45 Pa**

Hexagon Mesh Independence Study			
Mesh Size	**Mesh Element Counts**	**Result**	**%**
0.008	307894	61187.037	
0.006153846	523151	27630.74	121.4455241
0.004733728	**807555**	**27480.33**	**0.54733695**
0.003641329	1440572	27375.341	0.383516684

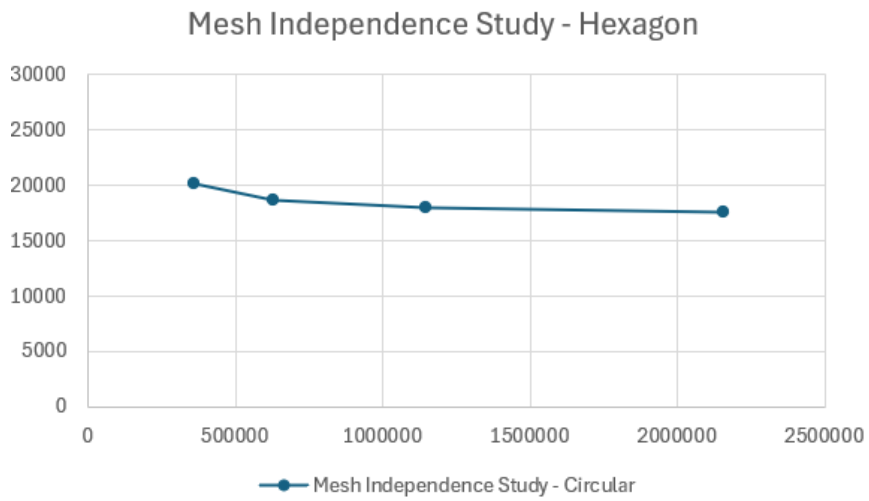


Figure 14. Hexagon Tube Mesh Independence Study

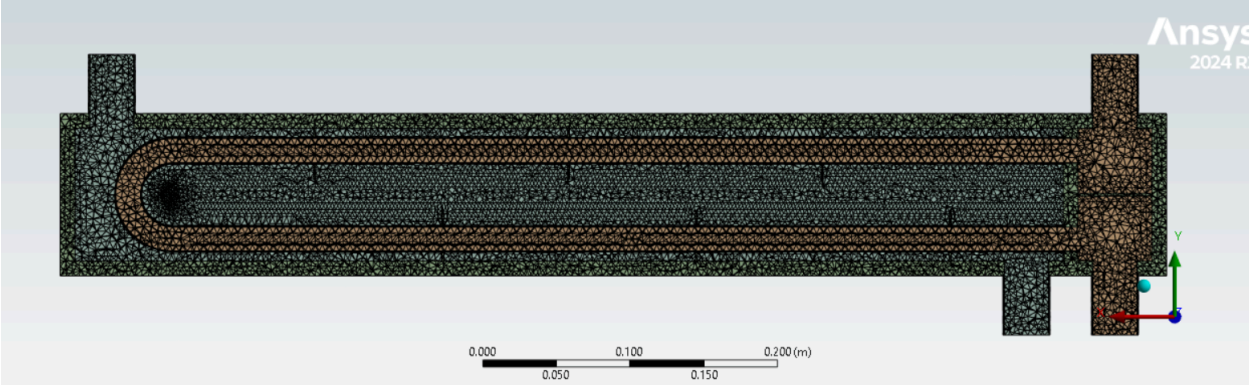


Figure 15. Hexagon Tube Mesh - 807,555 cells

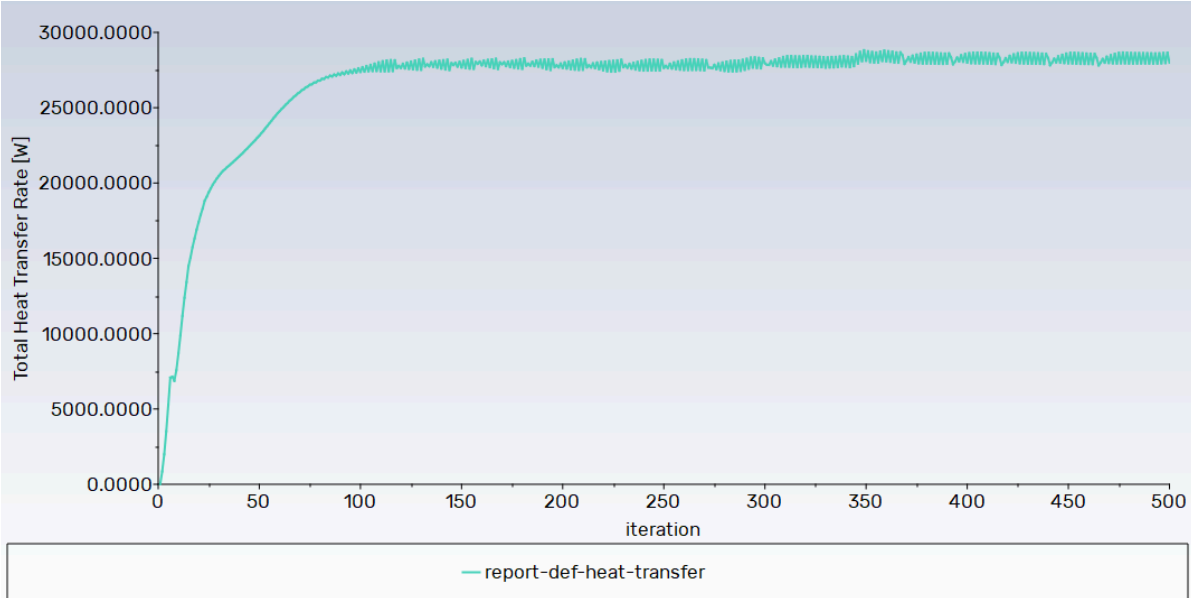


Figure 16. Heat Transfer Convergence Result - Hexagon @0deg

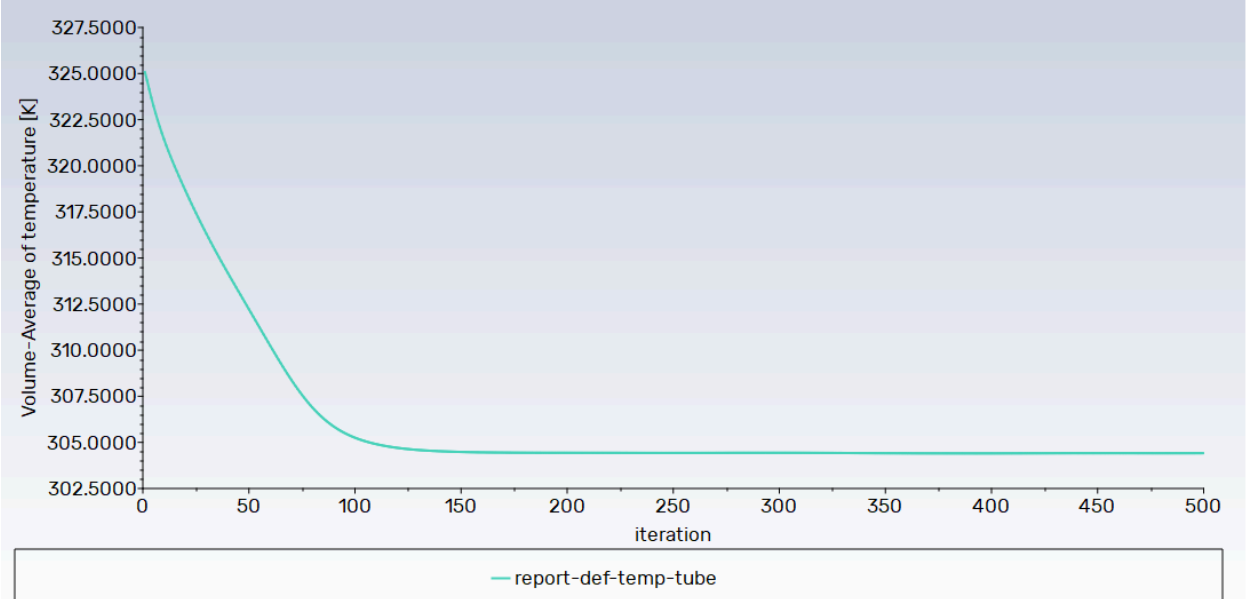


Figure 17. Temperature Tube Convergence Result - Hexagon @0deg

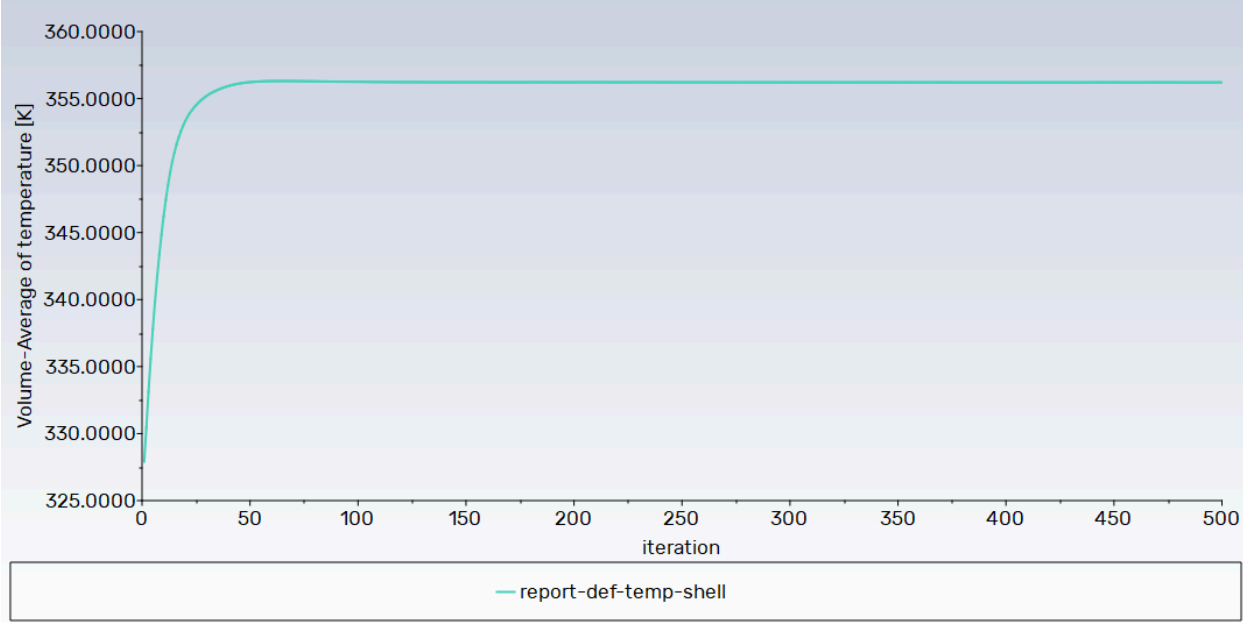


Figure 18. Temperature Shell Convergence Result - Hexagon @0deg

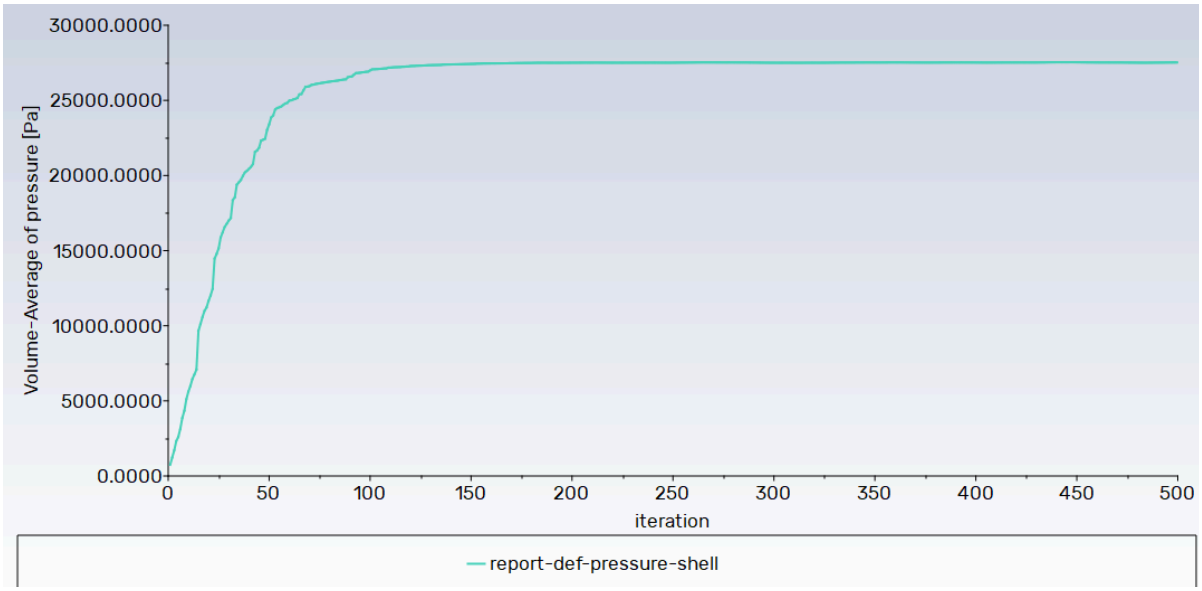


Figure 19. Pressure Shell Convergence Result - Hexagon @0deg

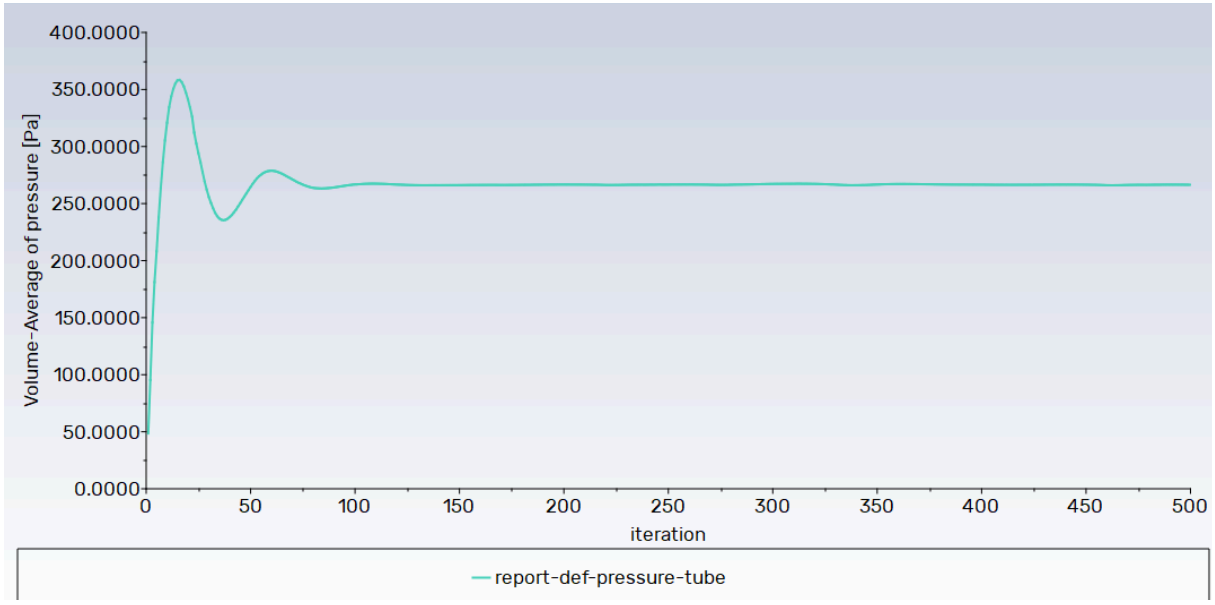


Figure 20. Pressure Tube Convergence Result - Hexagon @0deg

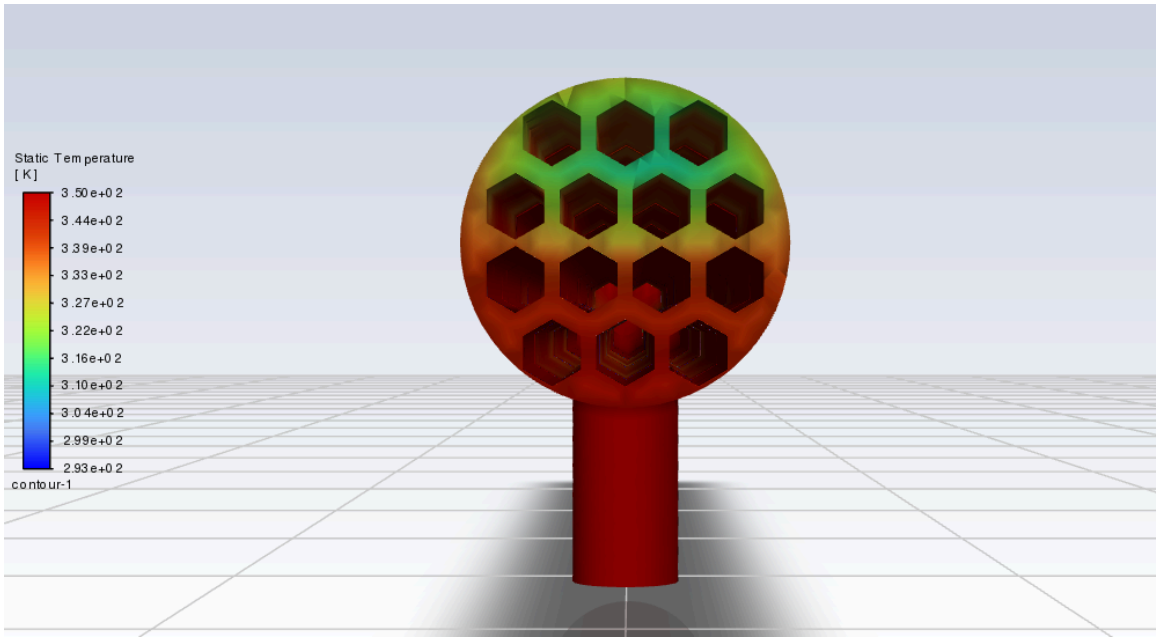


Figure 21. Post-Process View 1 - Hexagon @0deg

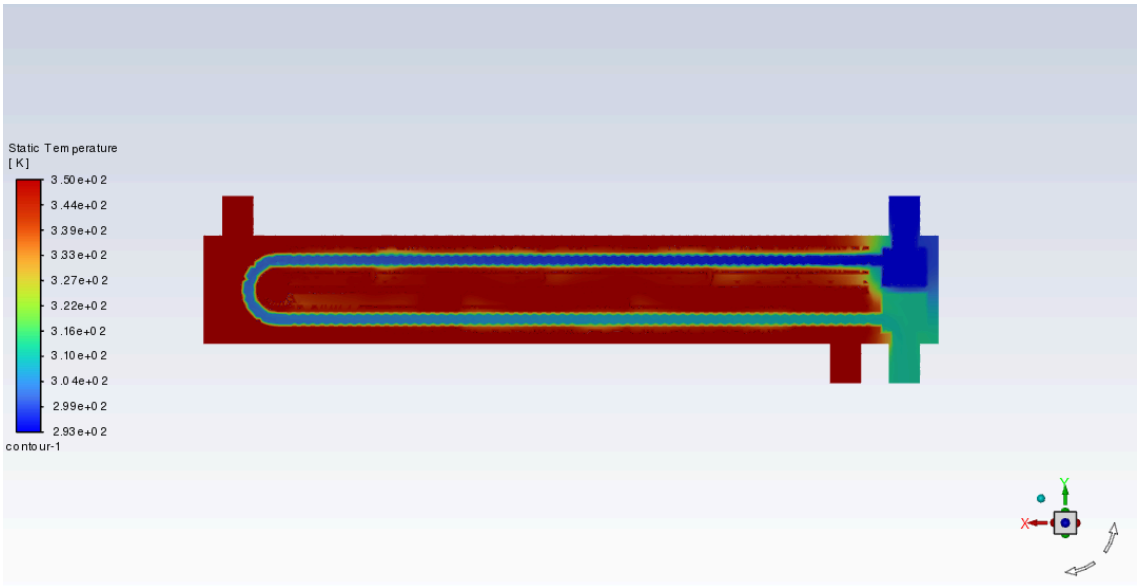


Figure 22. Post-Process View 2 - Hexagon @0deg

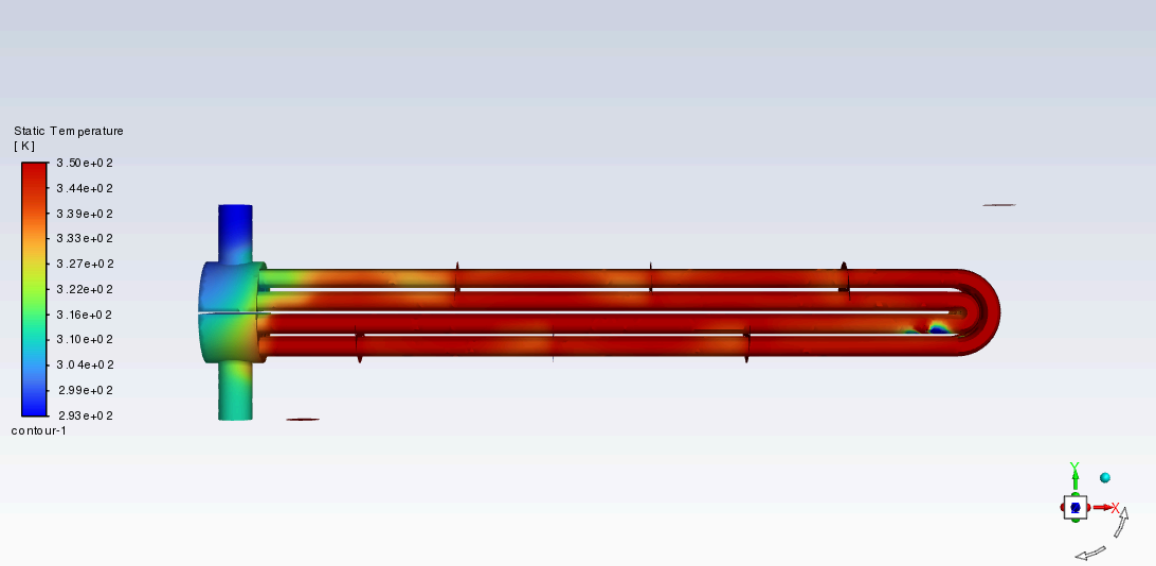


Figure 23. Post-Process View 3 - Hexagon @0deg

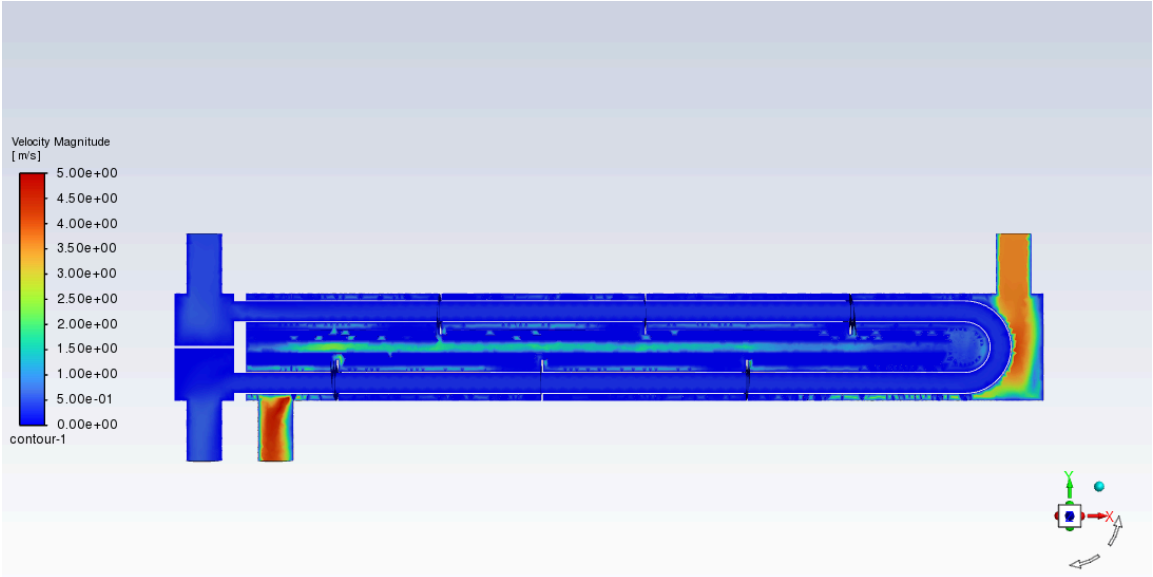


Figure 24. Post-Process View 4 - Hexagon @0deg

Hexagon @0deg Tube Results	
Heat Transfer	**25,579.57 W**
Temperature Tube	**304.41 K**

Hexagon @0deg Tube Results	
Temperature Shell	**356.20 K**
Pressure Shell	**27,525.46 Pa**
Pressure Tube	**266.47 Pa**

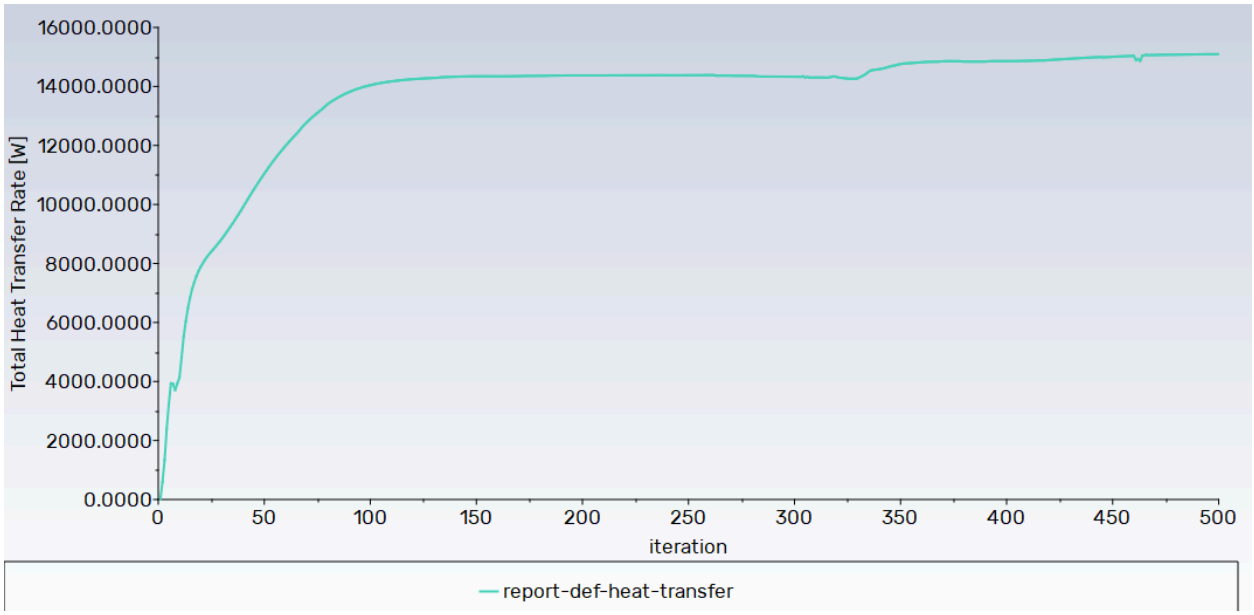


Figure 25. Heat Transfer Convergence Result - Hexagon@15deg

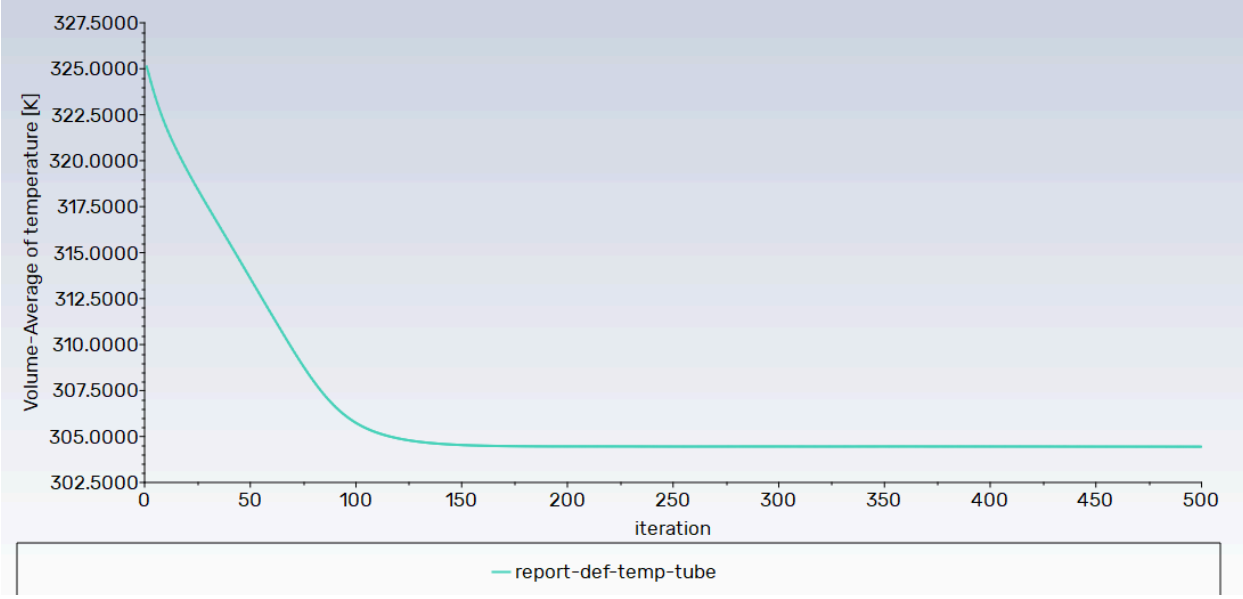


Figure 26. Temperature Tube Convergence Result - Hexagon@15deg

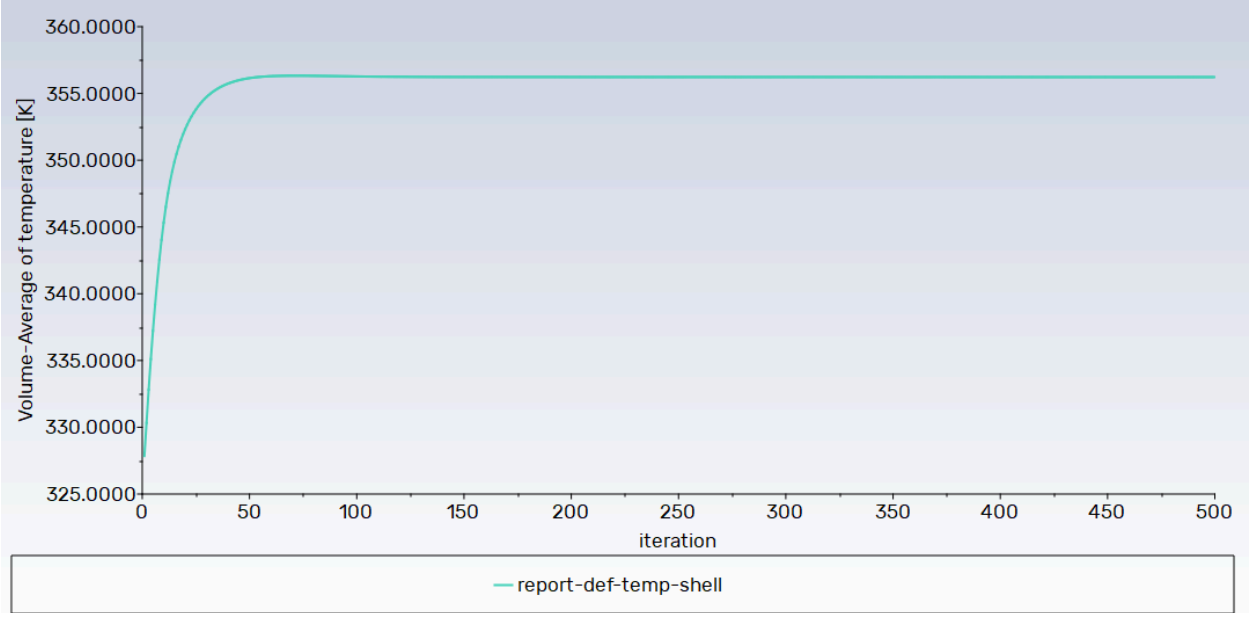


Figure 27. Temperature Shell Convergence Result - Hexagon@15deg

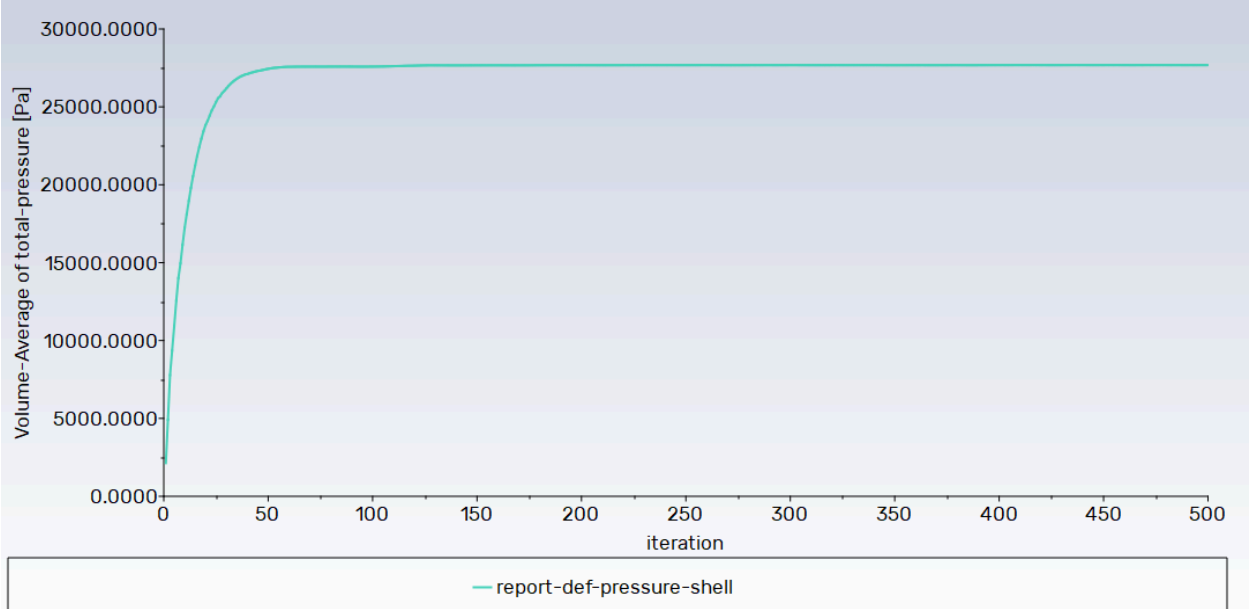


Figure 28. Pressure Shell Convergence Result - Hexagon@15deg

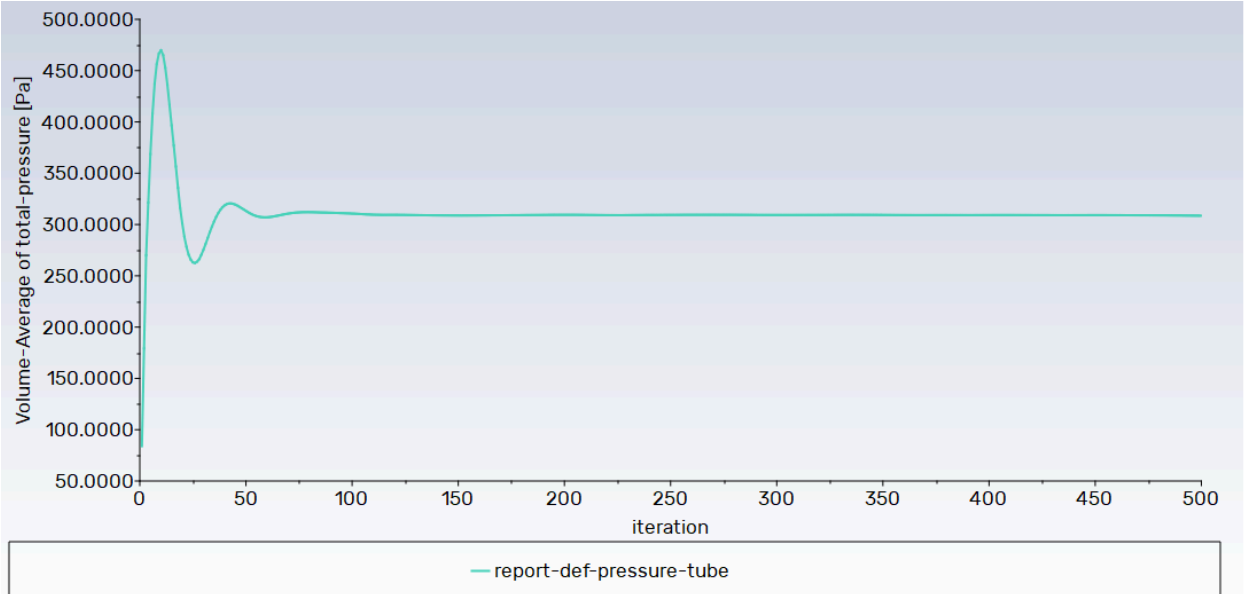


Figure 29. Pressure Tube Convergence Result - Hexagon@15deg

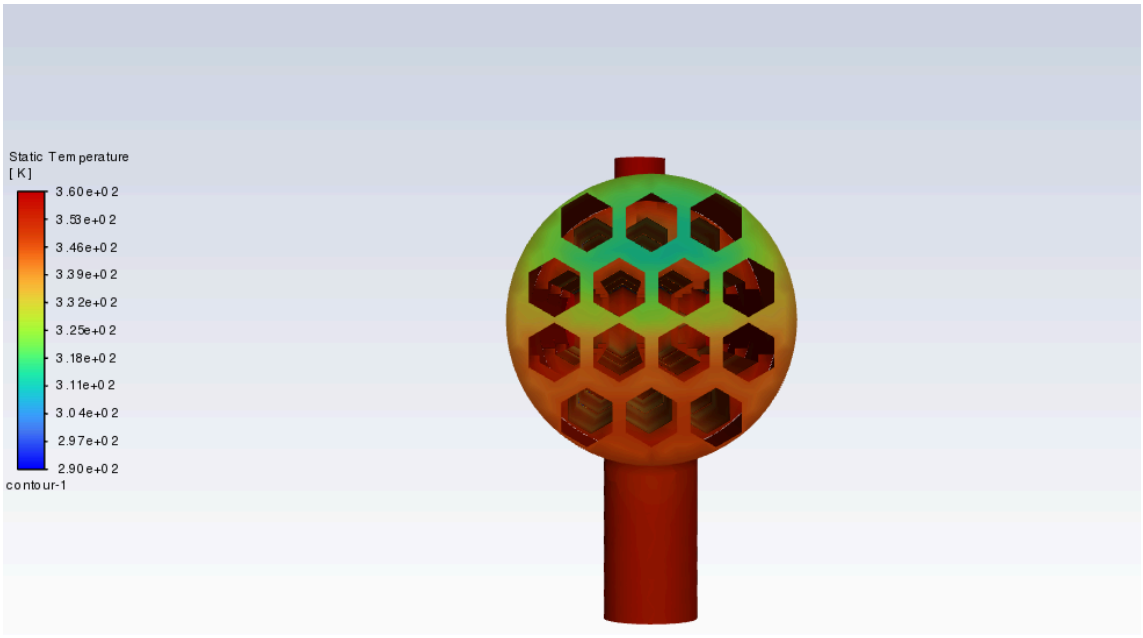


Figure 30. Post-Process View 1 - Hexagon@15deg

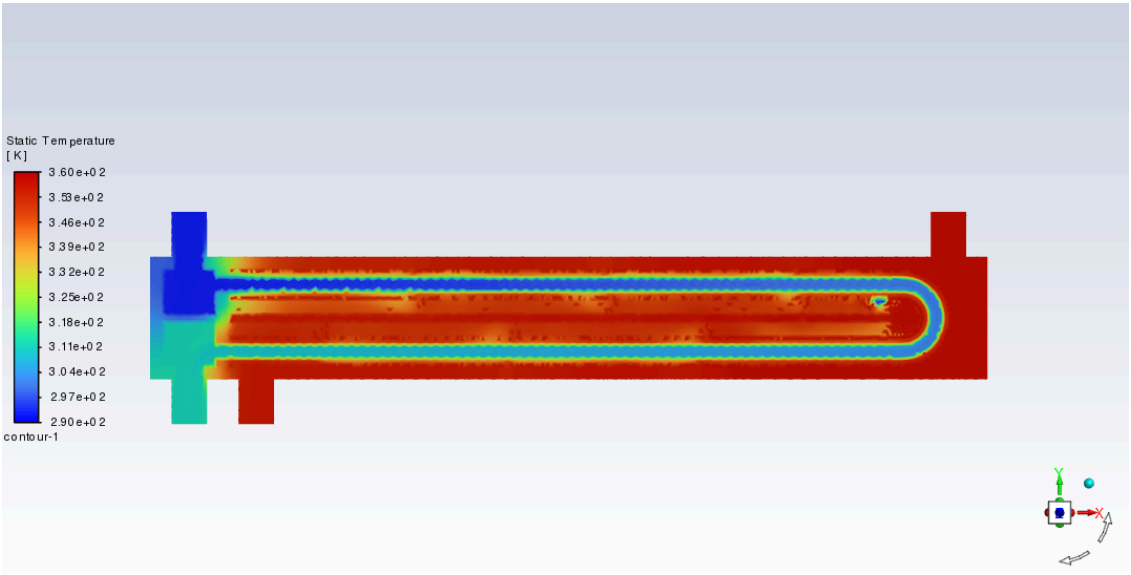


Figure 31. Post-Process View 2 - Hexagon@15deg

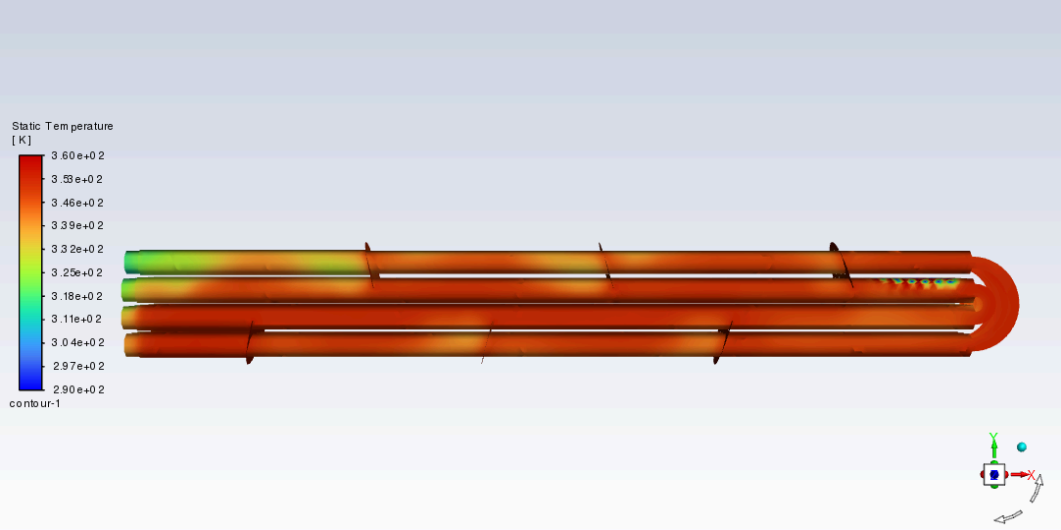


Figure 32. Post-Process View 3 - Hexagon@15deg

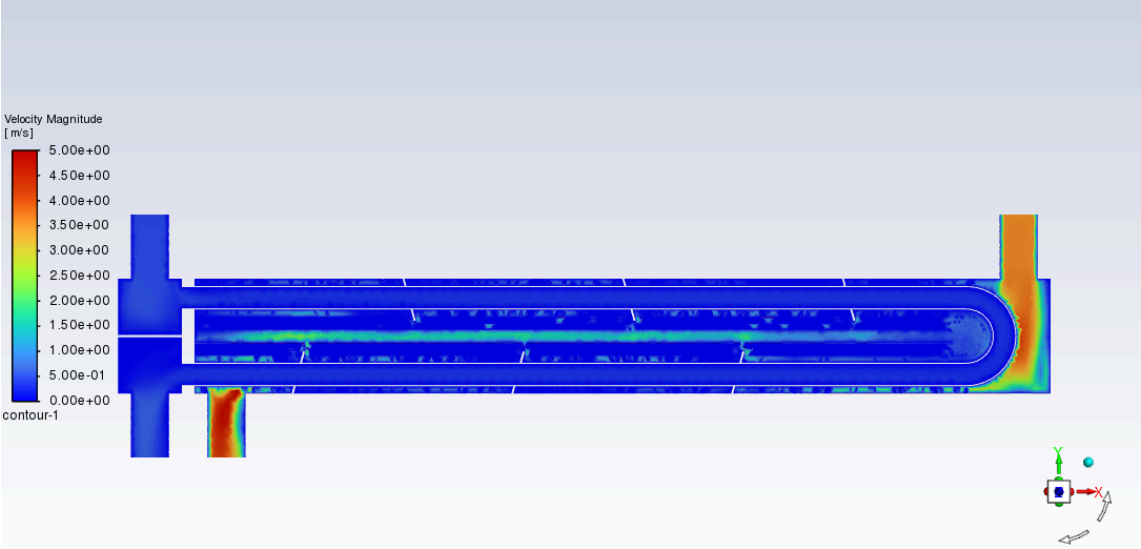


Figure 33. Post-Process View 4 - Hexagon@15deg

Hexagon @15deg Tube Results	
Heat Transfer	**15,095.12 W**
Temperature Tube	**304.41 K**
Temperature Shell	**356.20 K**

Hexagon @15deg Tube Results	
Pressure Shell	**27,525.46 Pa**
Pressure Tube	**266.47 Pa**

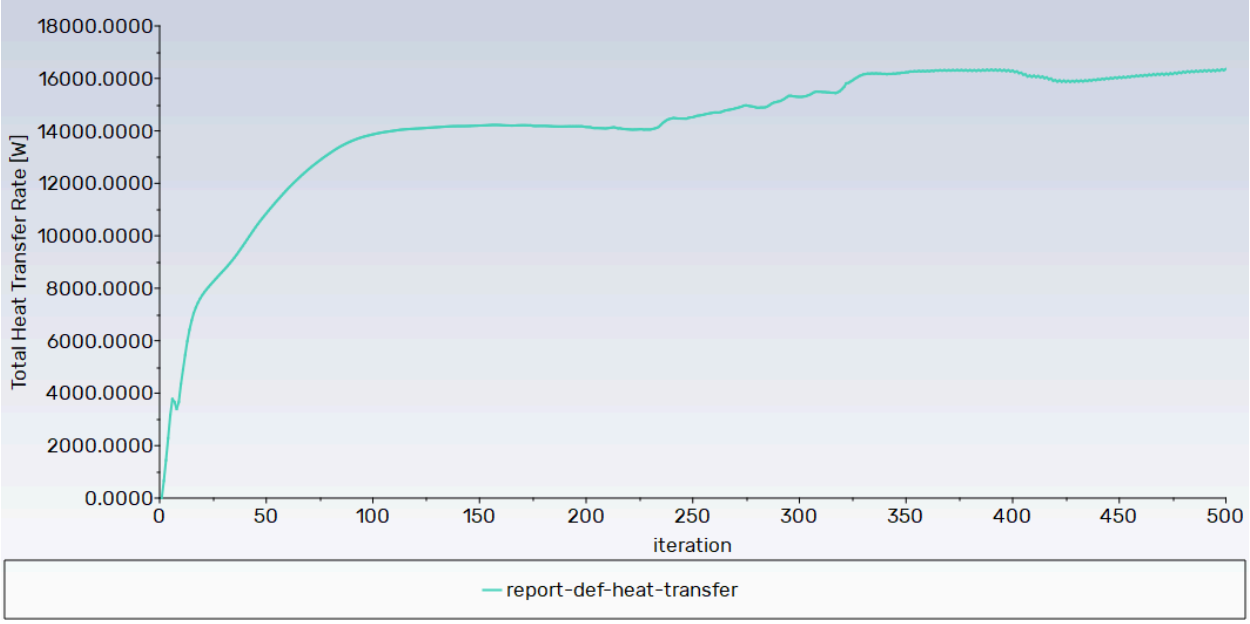


Figure 34. Heat Transfer Convergence Result - Hexagon@25deg

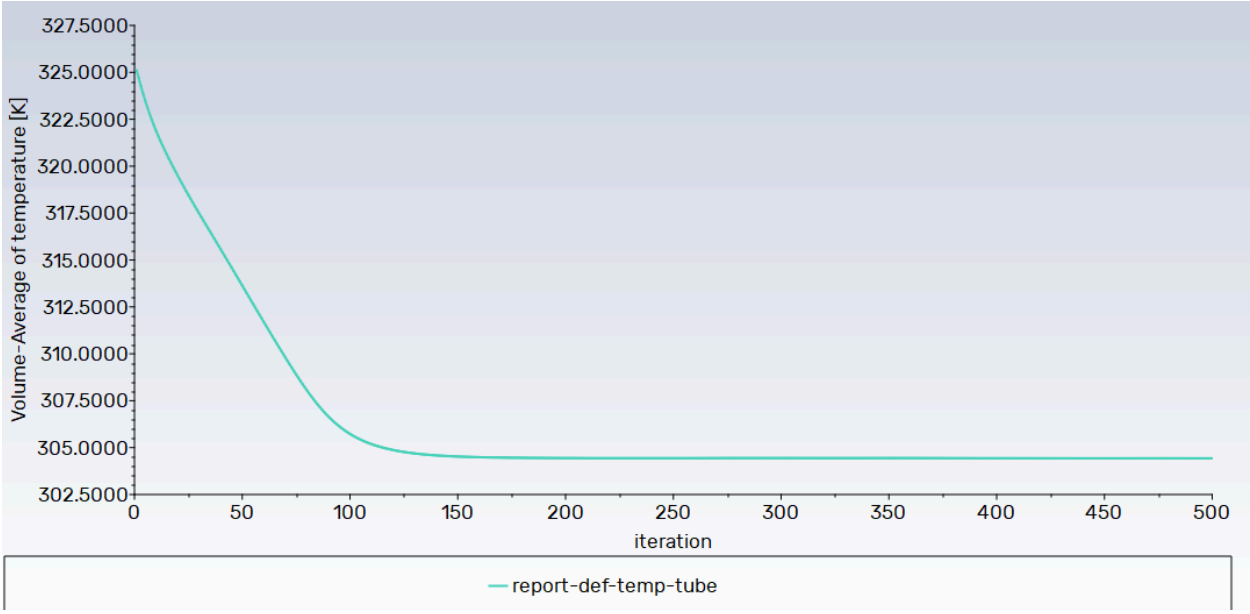


Figure 35. Temperature Tube Convergence Result - Hexagon@25deg

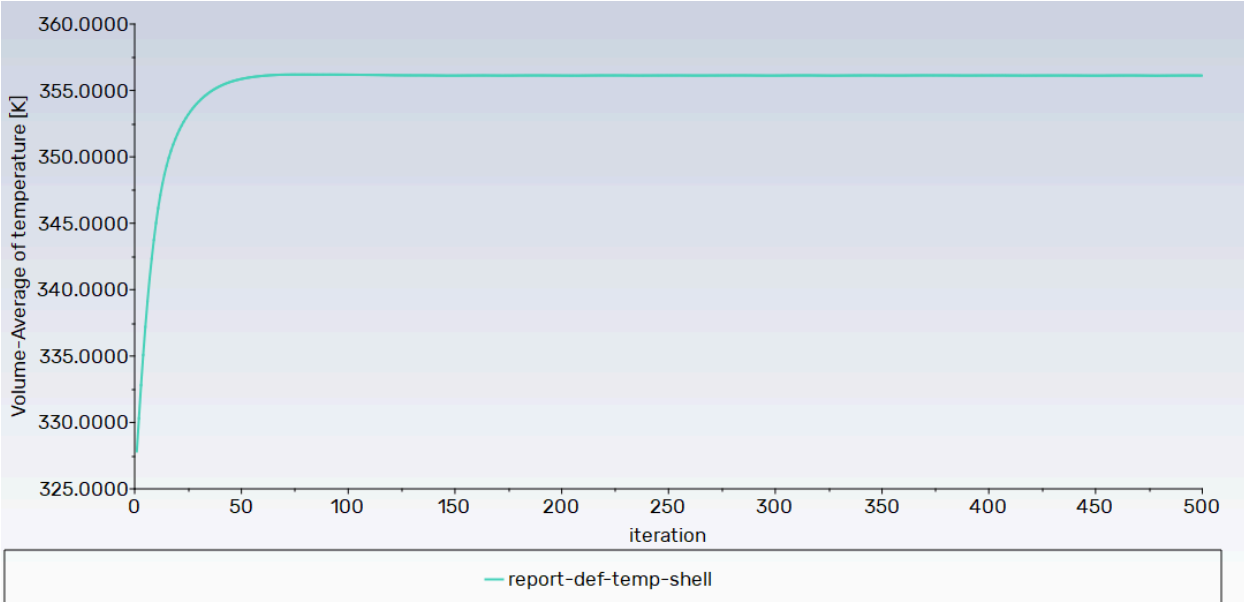


Figure 36. Temperature Shell Convergence Result - Hexagon@25deg

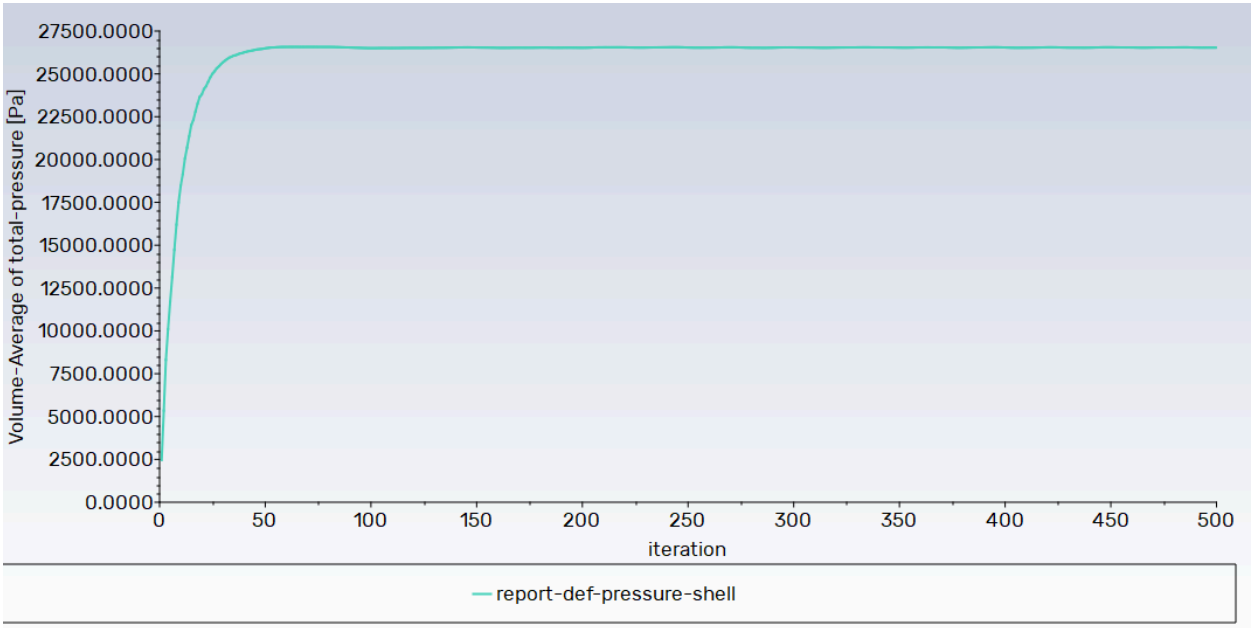


Figure 37. Pressure Shell Convergence Result - Hexagon@25deg

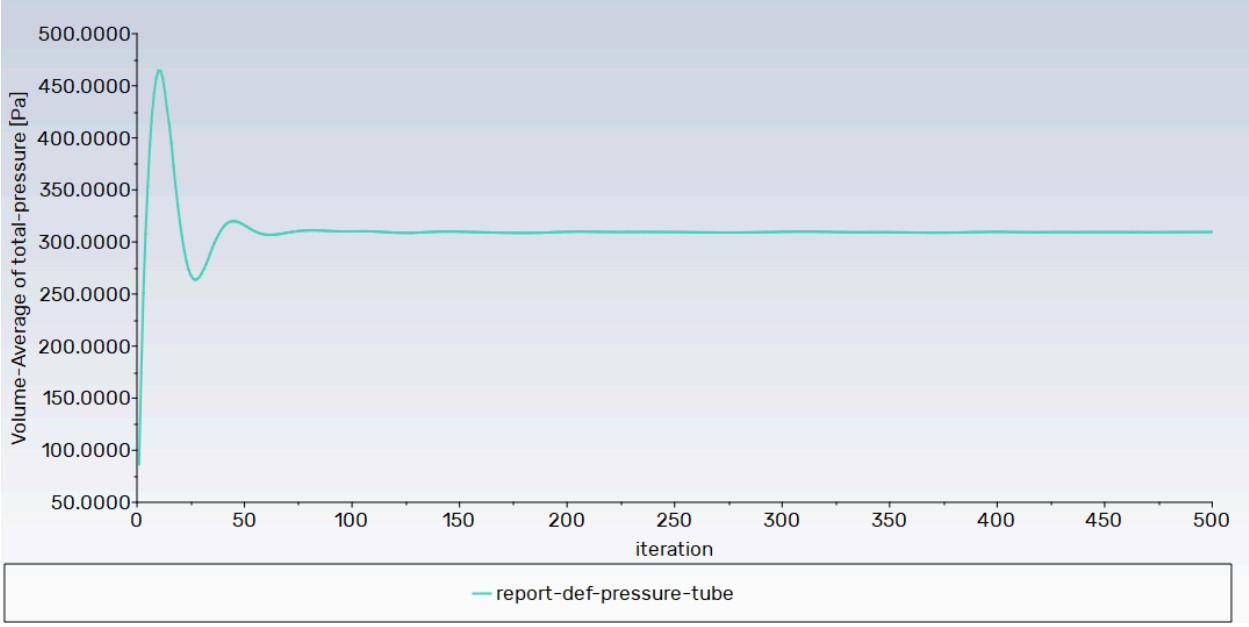


Figure 38. Pressure Tube Convergence Result - Hexagon@25deg

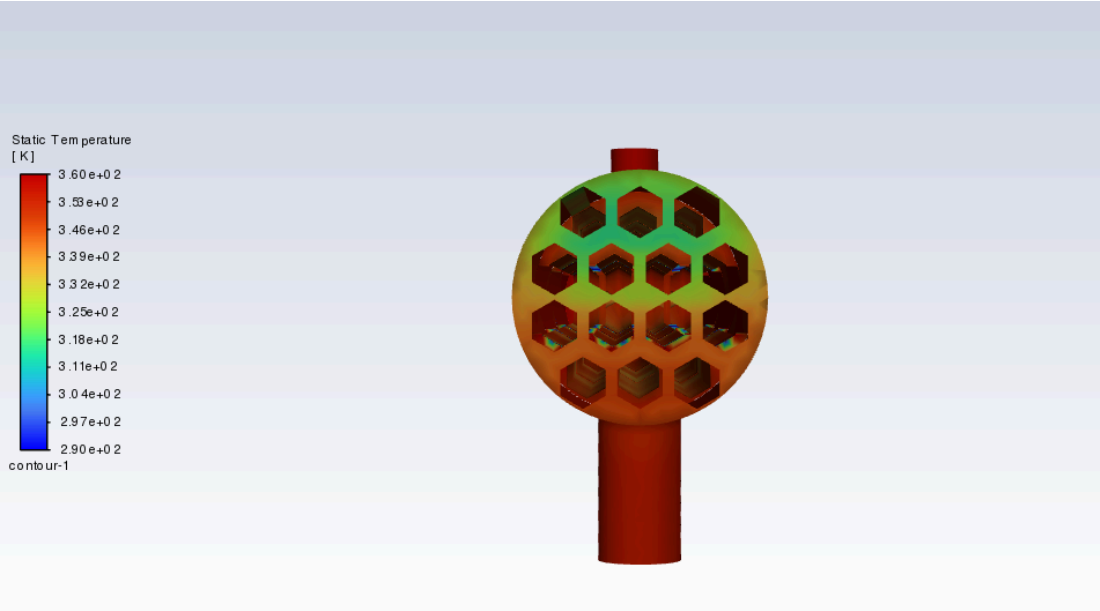


Figure 39. Post-Process View 1 - Hexagon@25deg

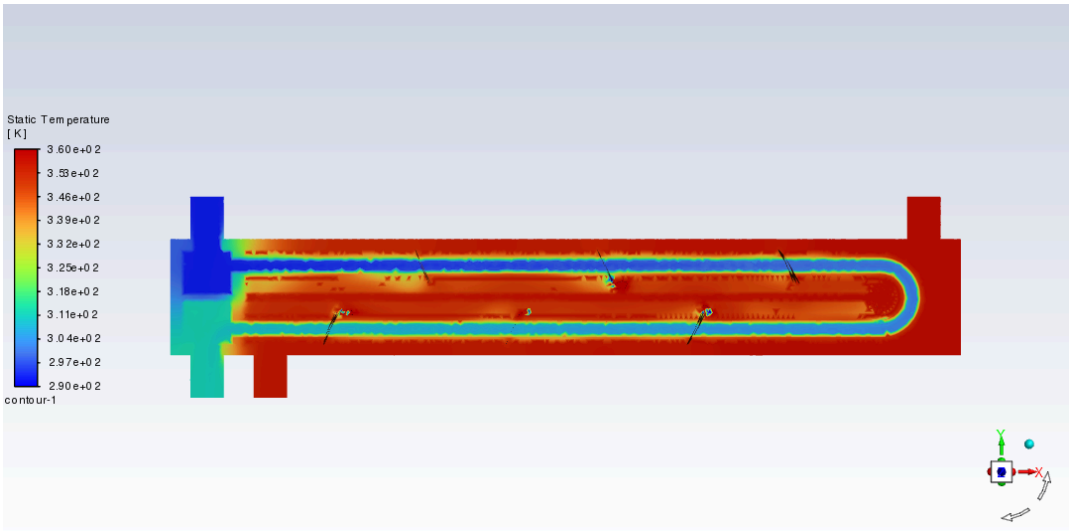


Figure 40. Post-Process View 2 - Hexagon@25deg

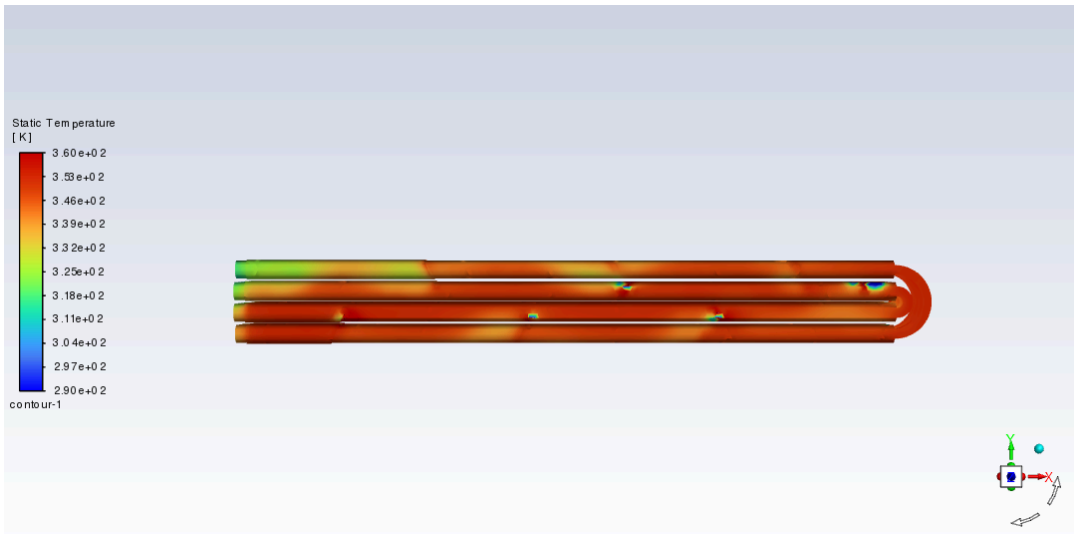


Figure 41. Post-Process View 3 - Hexagon@25deg

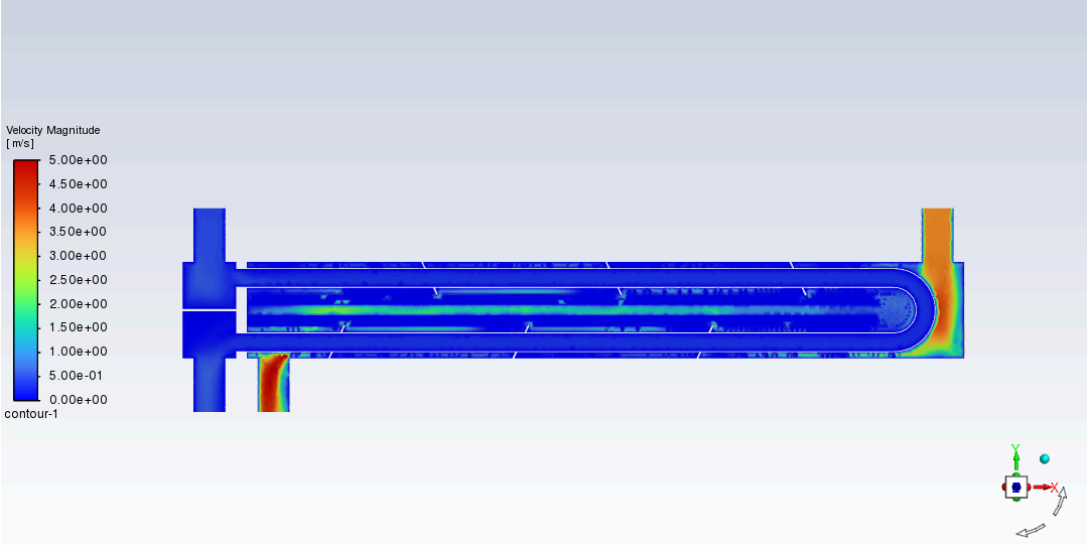


Figure 42. Post-Process View 4 - Hexagon@25deg

Hexagon @25deg Tube Results	
Heat Transfer	**16,358.42 W**
Temperature Tube	**304.42 K**
Temperature Shell	**356.12 K**
Pressure Shell	**26,533.46 Pa**
Pressure Tube	**309.62 Pa**

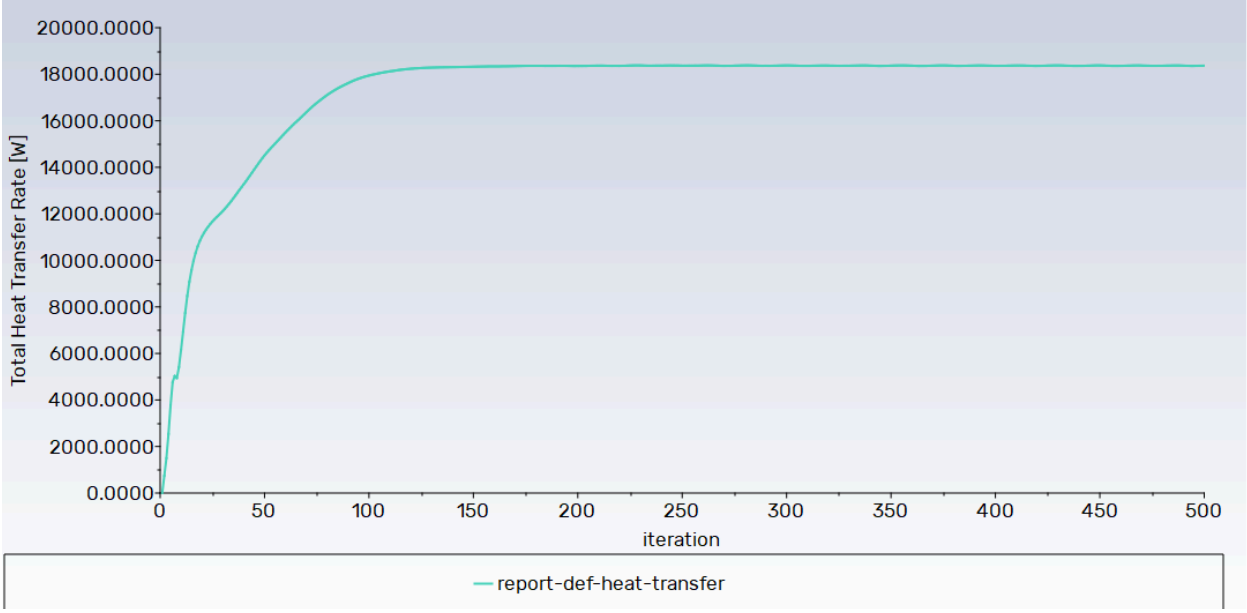


Figure 43. Heat Transfer Convergence Result - Hexagon@30deg

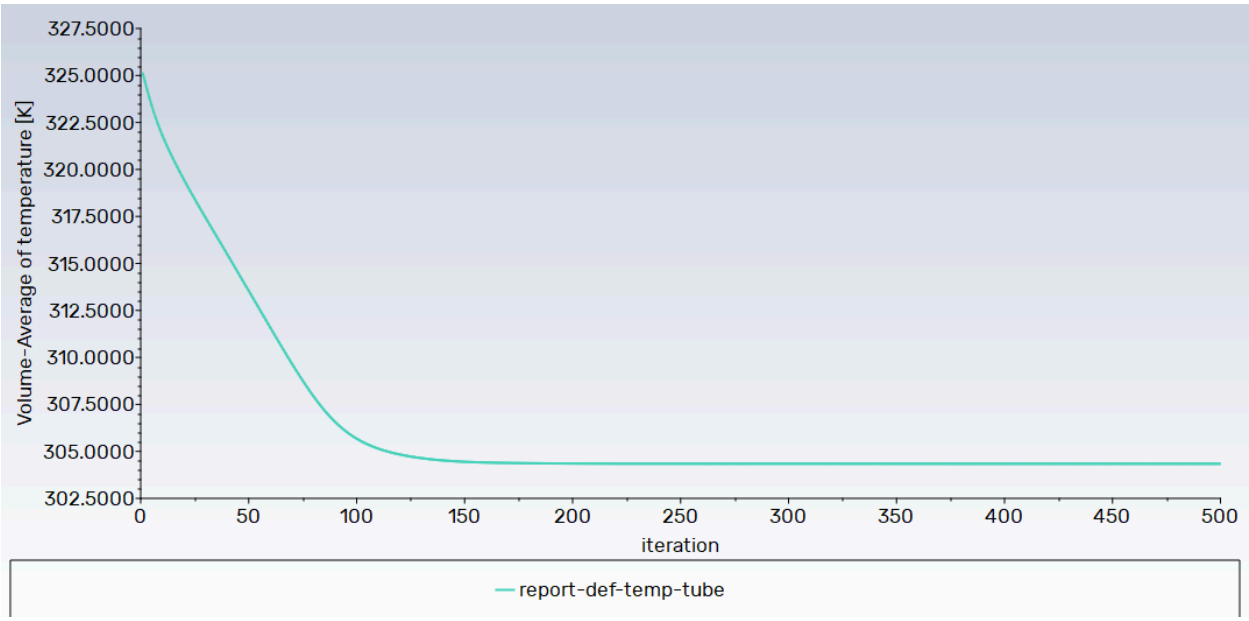


Figure 44. Temperature Tube Convergence Result - Hexagon@30deg

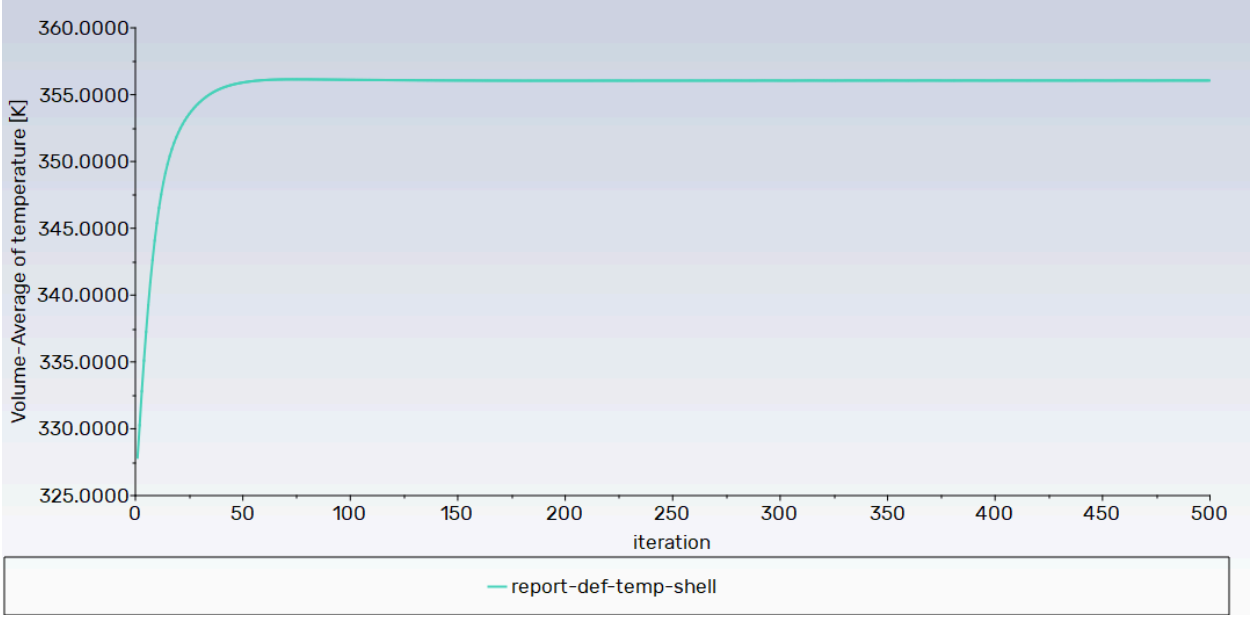


Figure 45. Temperature Shell Convergence Result - Hexagon@30deg

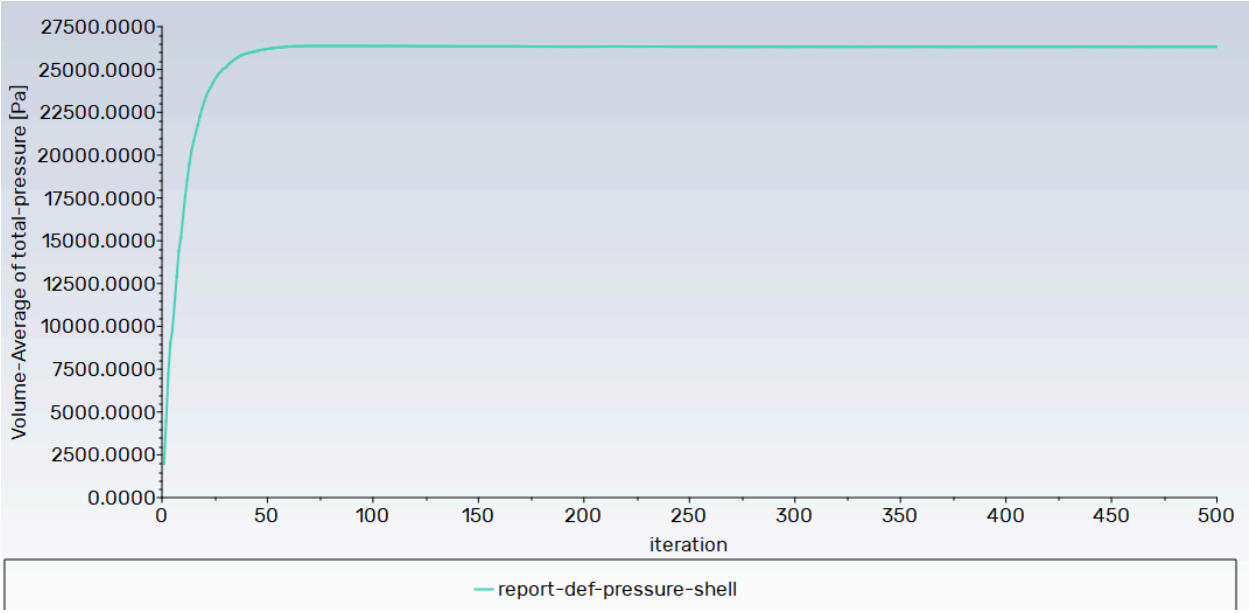


Figure 46. Pressure Shell Convergence Result - Hexagon@30deg

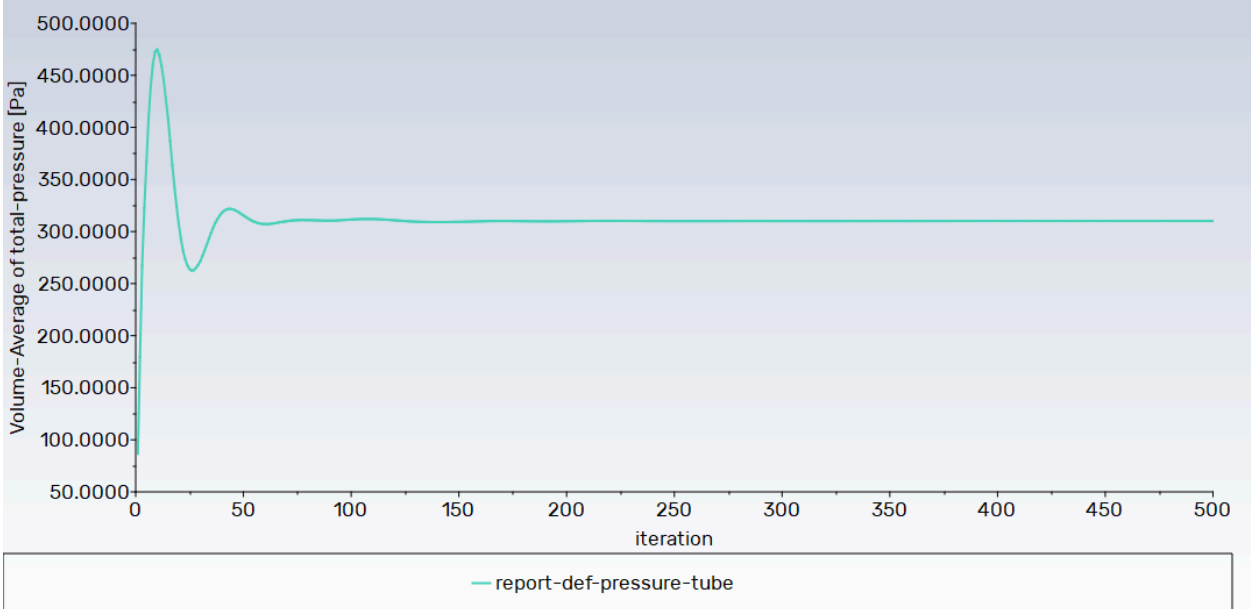


Figure 47. Pressure Tube Convergence Result - Hexagon@30deg

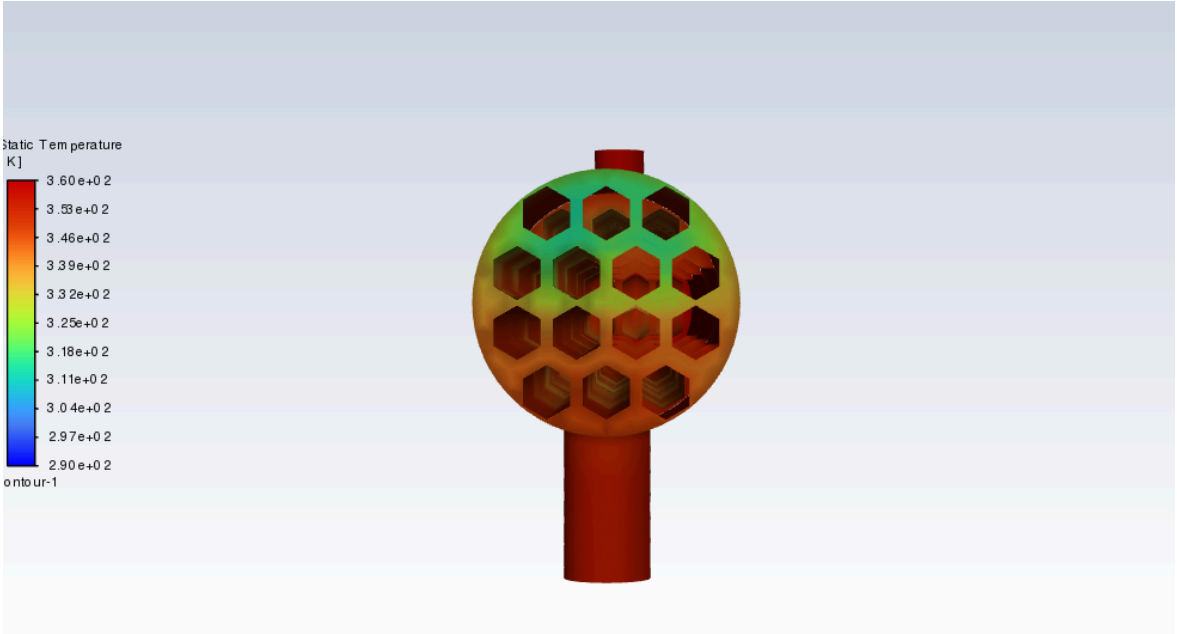


Figure 48. Post-Process View 1 - Hexagon@30deg

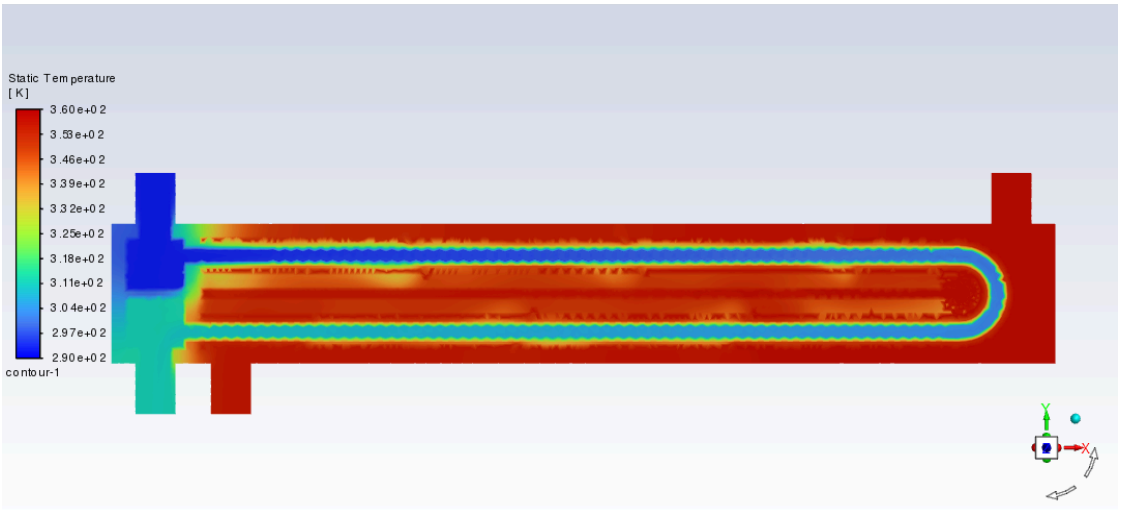


Figure 49. Post-Process View 2 - Hexagon@30deg

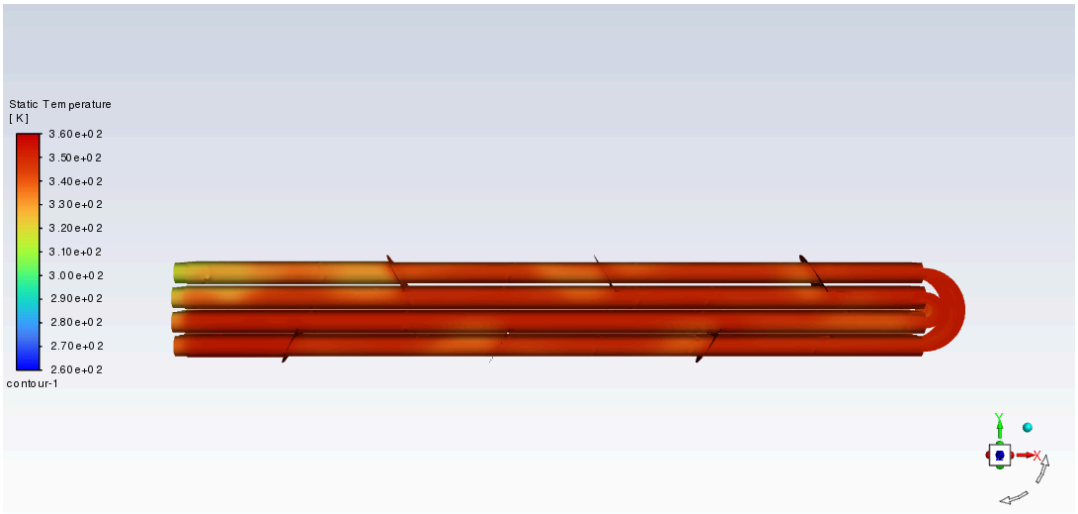


Figure 50. Post-Process View 3 - Hexagon@30deg

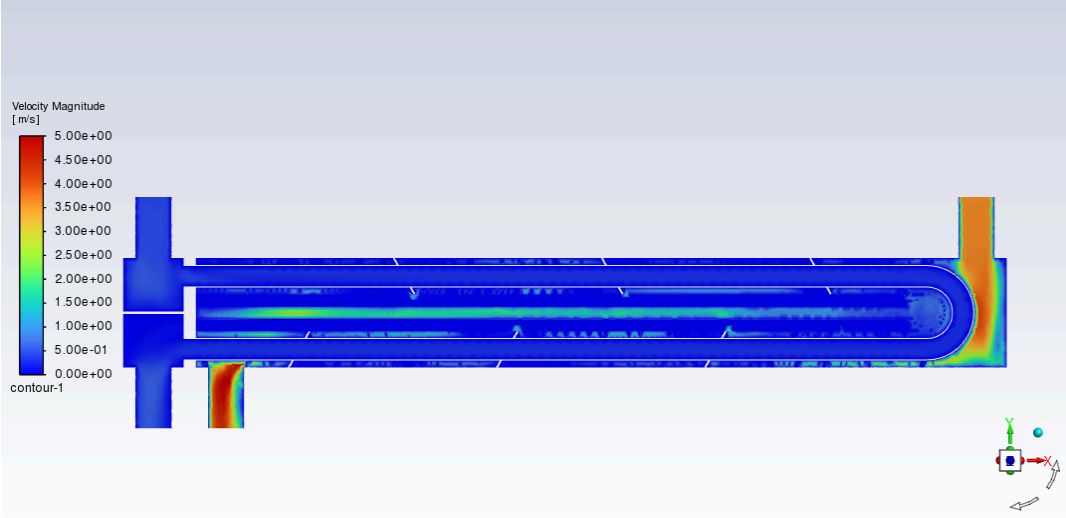


Figure 51. Post-Process View 4 - Hexagon@30deg

Hexagon @30deg Tube Results	
Heat Transfer	**18,368.35 W**
Temperature Tube	**304.34 K**
Temperature Shell	**356.05 K**
Pressure Shell	**26,323.10 Pa**
Pressure Tube	**310.14 Pa**

3. Summary

Hexagon @30deg Tube Results					
	Circular	Hexagon @0deg	Hexagon @15deg	Hexagon @25deg	Hexagon @30deg
Heat Transfer	25,579.57 W	25,579.57 W	15,095.12 W	16,358.42 W	18,368.35 W
Temperature Tube	303.78 K	304.41 K	304.41 K	304.42 K	304.34 K
Temperature Shell	356.34 K	356.20 K	356.20 K	356.12 K	356.05 K
Pressure Shell	18,061.43 Pa	27,525.46 Pa	27,525.46 Pa	26,533.46 Pa	26,323.10 Pa
Pressure Tube	353.45 Pa	266.47 Pa	266.47 Pa	309.62 Pa	310.14 Pa

NOTHING FOLLOWS